AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Engineering

Lab Report

Course Title:	COMPUTER NETWORKS		
Course Code:	COE3206	**Section:**	**C**
Semester:	Summer 2025-26	**Degree Program:**	BSc in CSE
Course Teacher:	**DR. RAJARSHI ROY CHOWDHURY**		

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2.	**Md Tauhidul Islam Siam**	23-53927-3	Cse	
3.	**MD.SAIDUR RAHMAN ROHAN**	23-53891-3	Cse	

LAB-1

Required Devices:-

- 5 Laptops
- 8 Pcs
- 3 Hub-PT
- 1 Server-PT
- 2 Copper Cross-Over Cable
- 13 Copper Cable

1. Step-1:



Add 8 Pc and 5 Laptop and 1 Server from this part .

2. Step-2:-



Add 3 Hub-PT from this part

-

3. Step-3:-



From this part we select cable to connect the component . In this lab we use 2 types of Cable. One is Copper Cross-Over Cable's and another is Normal Copper Cable's.

For Hub and laptop or pc connection And Hub to Server connection we use **Copper Cables** . Like this:-

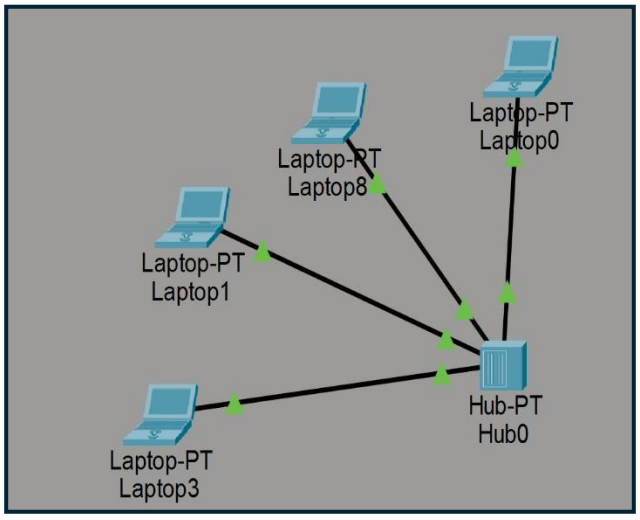


Fig:1 :- Hub to Laptop connection.

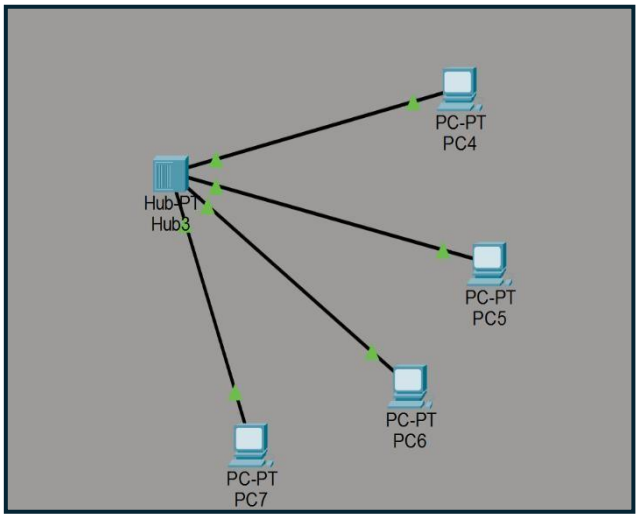


Fig:2- Hub to Pc connection.

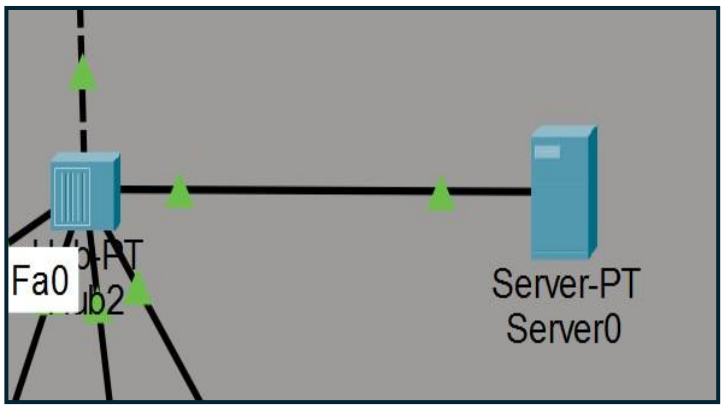


Fig:3:-Hub to Server connection.

For Hub to Hub Connection We use **Copper Cross-Over Cable's.** Like this:-

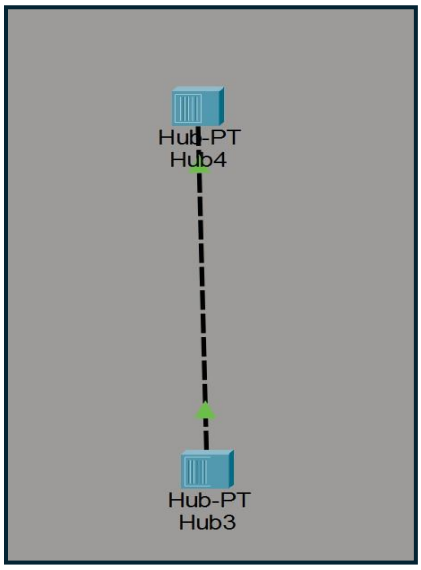
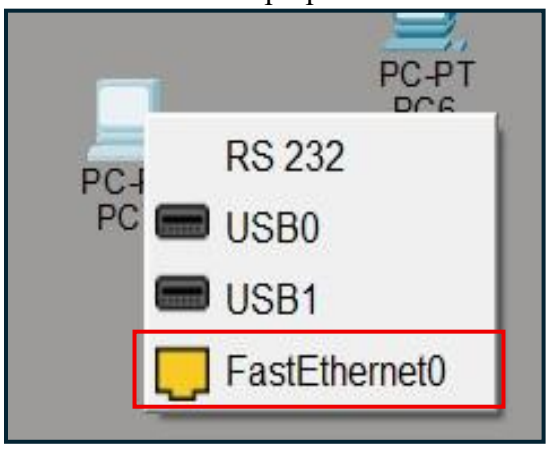
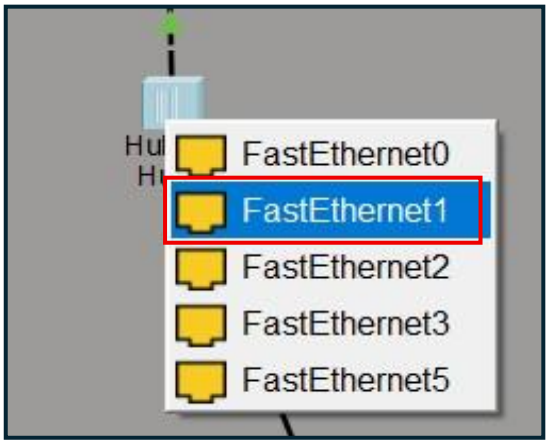


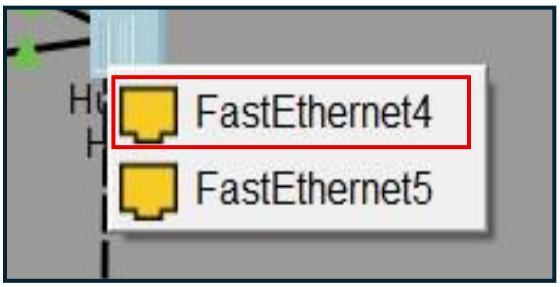
Fig:4:- Hub to Hub Connection.

For Hub To PC or laptop Connection We use FastEthernet Port:-





For Hub-to-Hub Connection We also use Ethernet Port:



After Connection The Component My Final Design is :

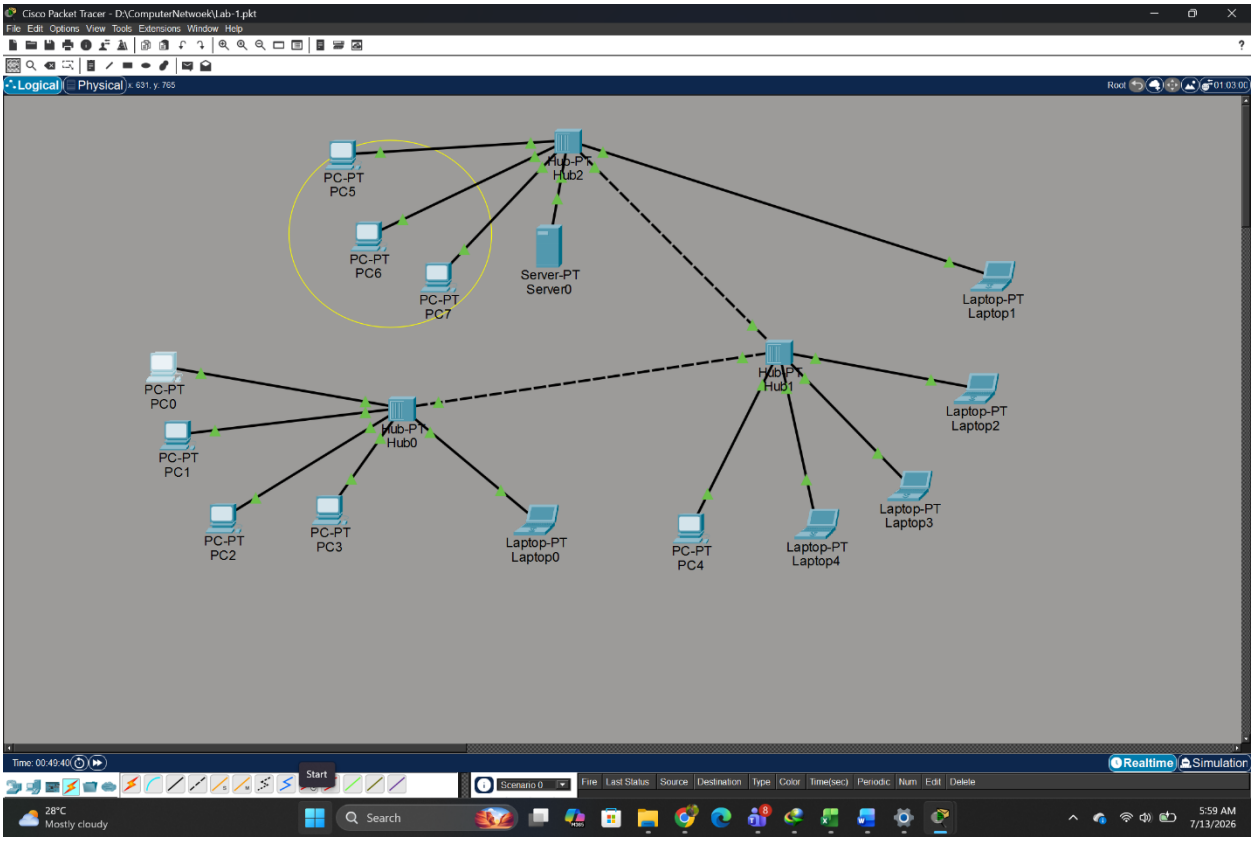
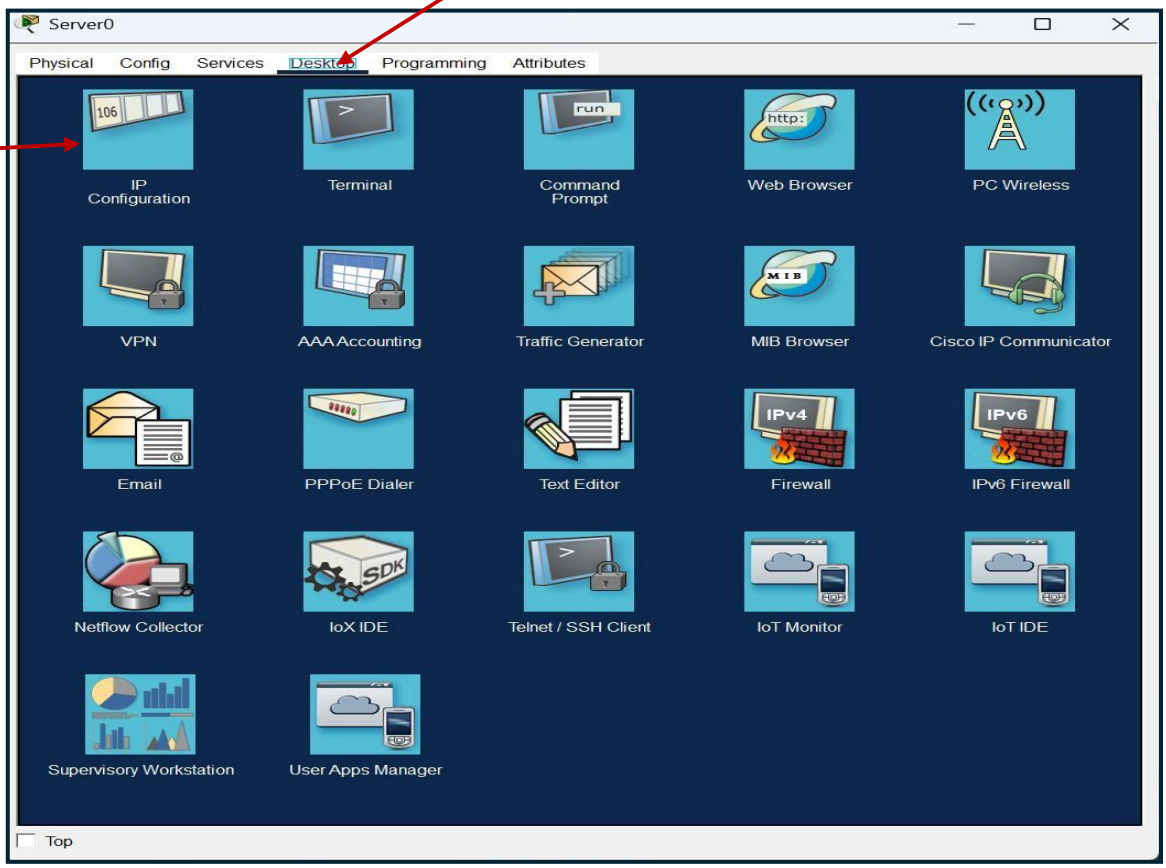


Fig:6:- Final Design.

4. Step:4:- Now, We setup Server

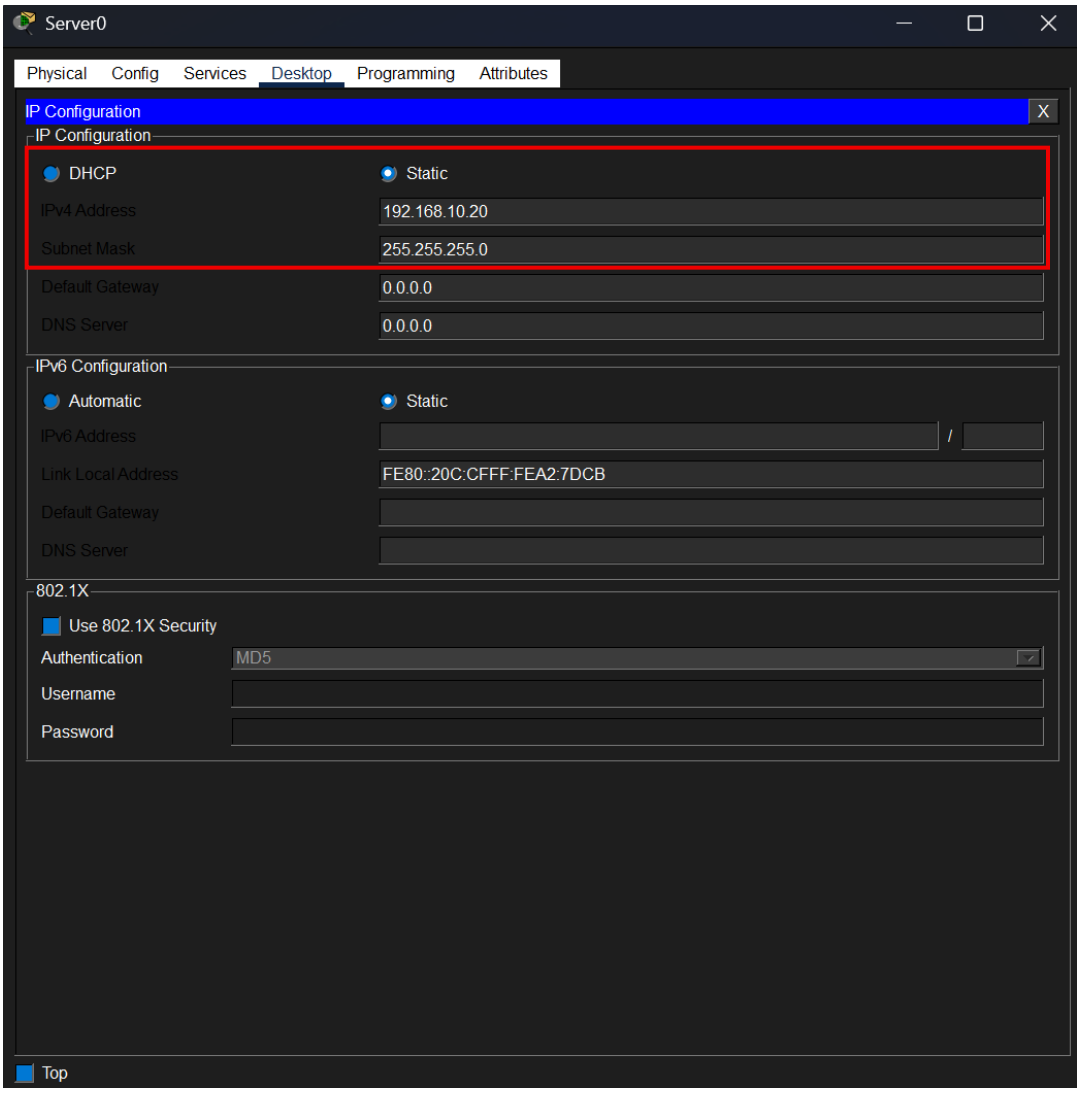
Click Server → Desktop → IP Configuration



Then,

Select IP Address Static

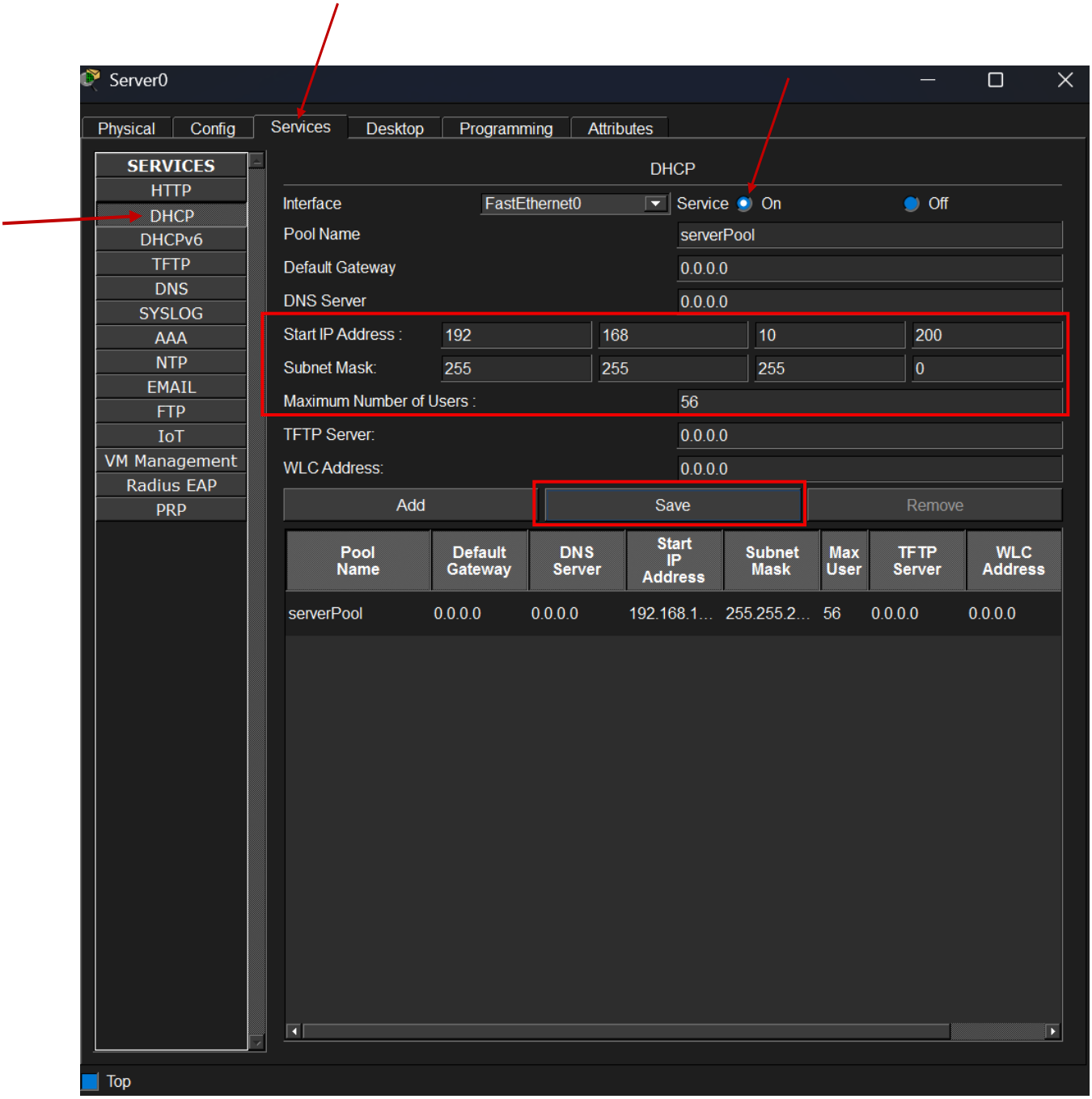
IPv4 Address:- 192.168.10.20
Subnet Mask :255.255.255.



Then Go to Service → DHCP

- DHCP Service On
- Start IP Address:192.168.10.20
- Subnet Mask :255.255.255.0

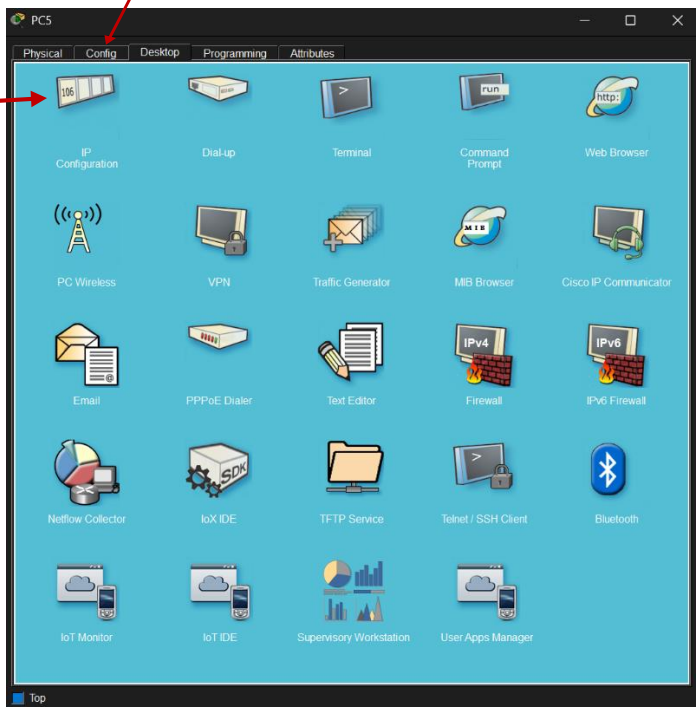
Then Save the DHCP Sitting.

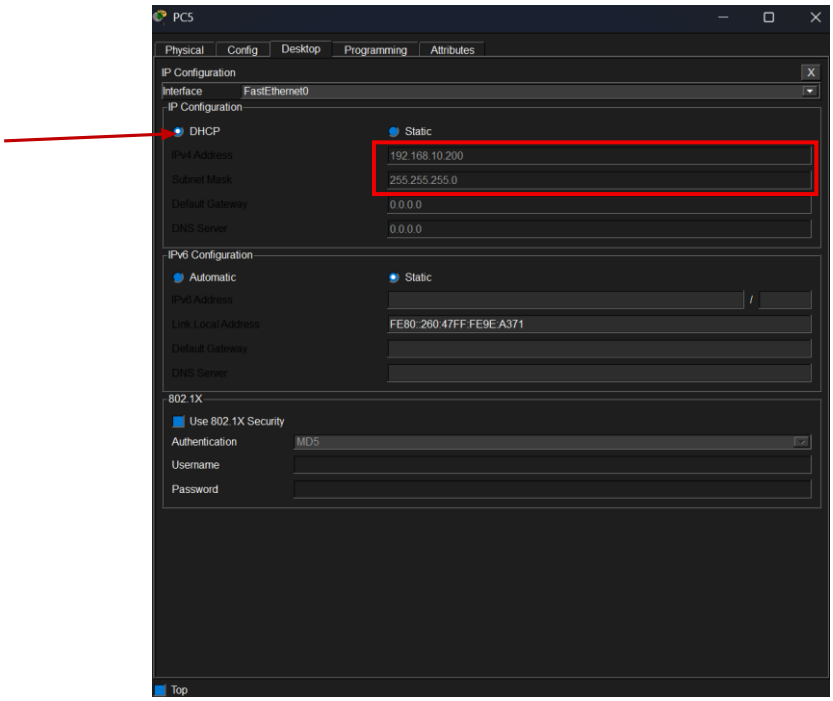


5. Step:5: -

Now, we setup Three Pc Like this way: -

Click PC → Desktop → IP Configuration ->Static->IPv4 (as like picture below)



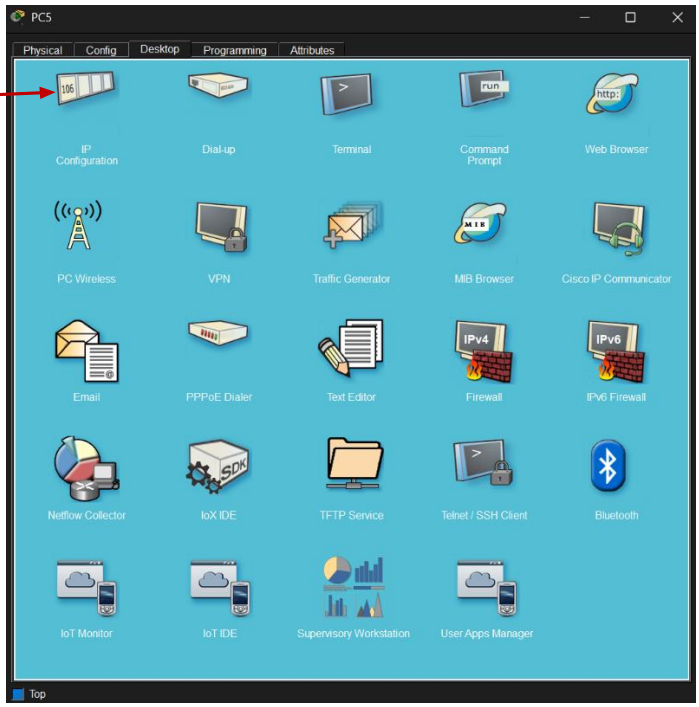


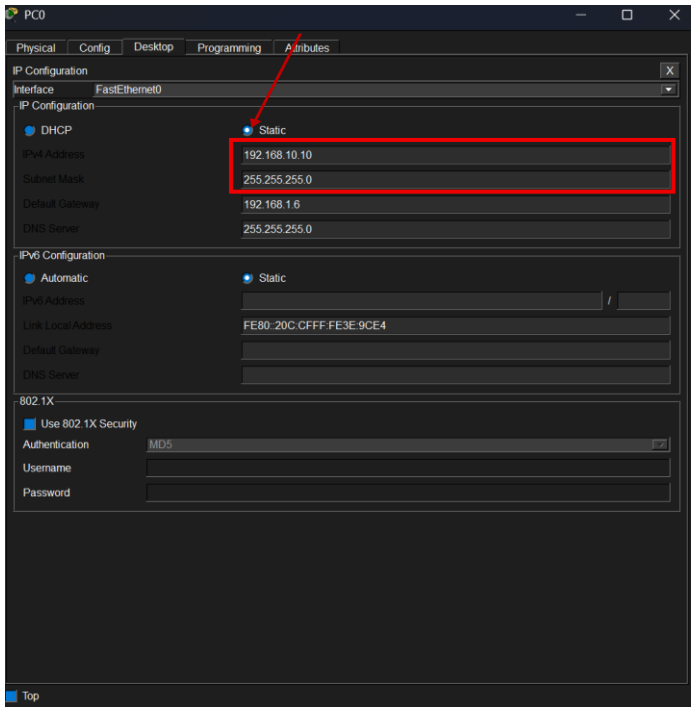
Then , Select Ip Configuration DHCP

There are only three PCs connected to **DHCP**, server it will assign IP addresses sequentially starting from **192.168.10.200**.

6. Step:6: - Other 5 pcs and 5 laptop are assign by manual IP addresses like

Click PC → Desktop → IP Configuration ->Static->IPv4 (as like picture below)





Select Ip configuration on static.

7.Step:7: - Ip Addressing Table

DHCP SERVER

Device Name	IP Address	Subnet Mask
PC5	192.168.10.200	255.255.255.0
PC6	192.168.10.201	255.255.255.0
PC7	192.168.10.202	255.255.255.0

Static Manual Addressing

Device Name	IP Address	Subnet Mask
PC0	192.168.10.10	255.255.255.0
PC1	192.168.10.11	255.255.255.0
PC2	192.168.10.12	255.255.255.0
PC3	192.168.10.13	255.255.255.0
Laptop0	192.168.10.14	255.255.255.0
PC4	192.168.10.15	255.255.255.0
Laptop4	192.168.10.16	255.255.255.0
Laptop3	192.168.10.17	255.255.255.0
Laptop2	192.168.10.18	255.255.255.0
Laptop1	192.168.10.19	255.255.255.0

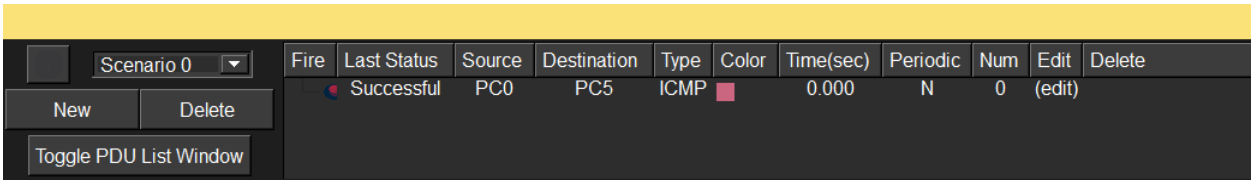
8.Step:8: - Add a Simple PDU (Packet) 1.Select Add Simple PDU (envelope icon).



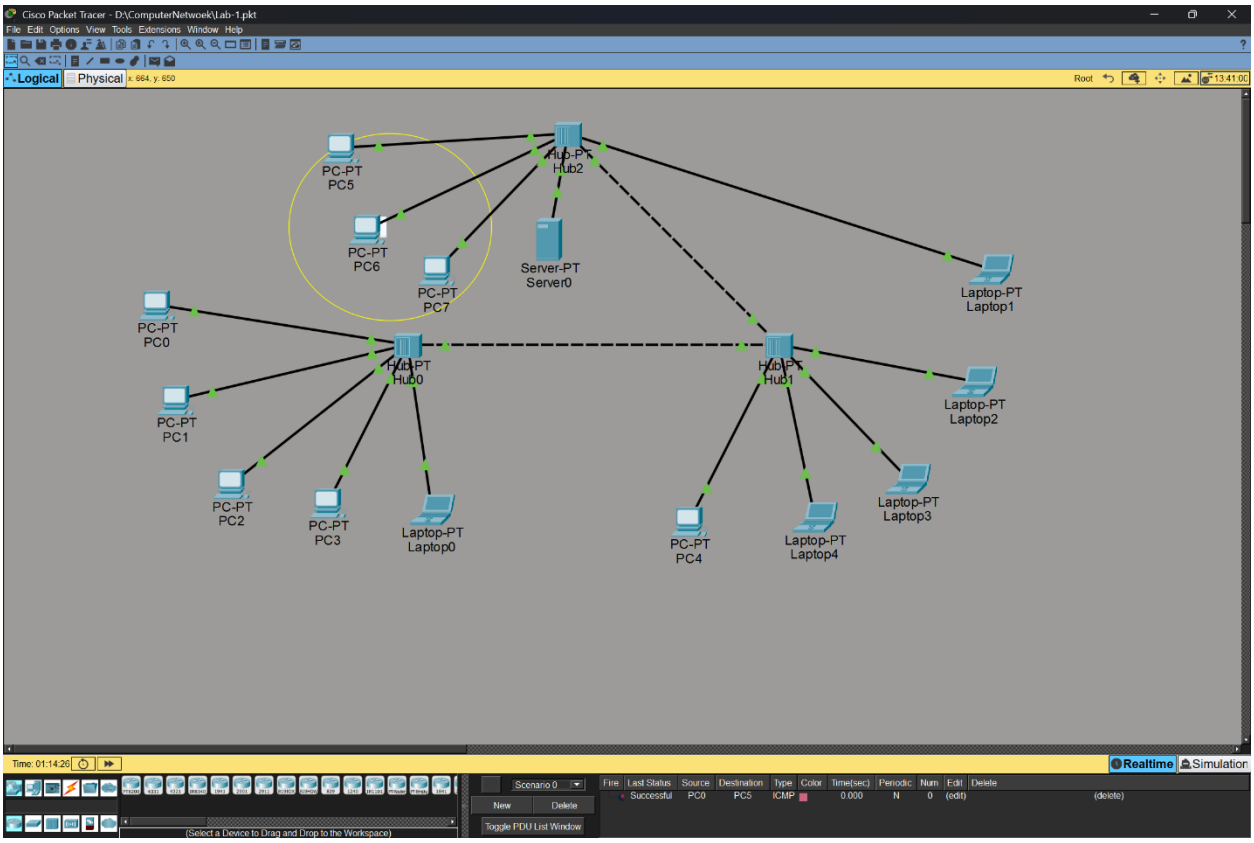
Click on PC-0, then on PC-5 as For Packet Transfer . Observe the envelope (packet) moving through: Pc0→ Hub0 →Hub1→ Hub 2→ Server → PC-5

- The Event List showed:

. Successful → if the packet reaches PC-5 correctly.



Final picture of completed lab task-1



LAB-2

Required Devices:-

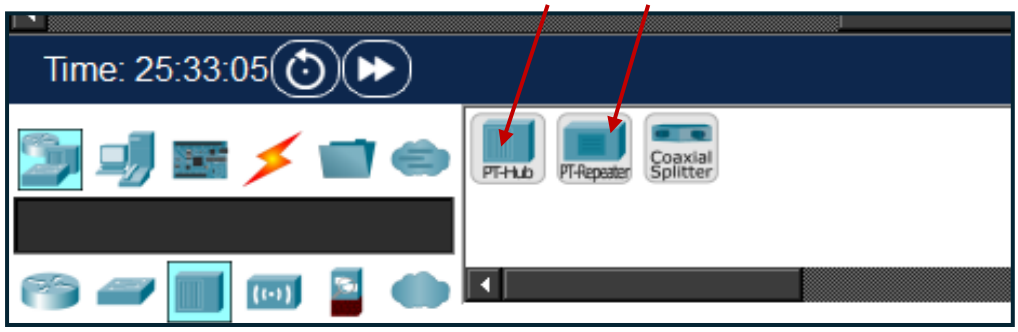
- 10 Laptops
- 10 Pcs
- 5 Hub-PT
- 1 Server-PT
- 1 Repeater
- 5 Copper Cross-Over Cable
- 21 Copper Cable

1. Step-1:



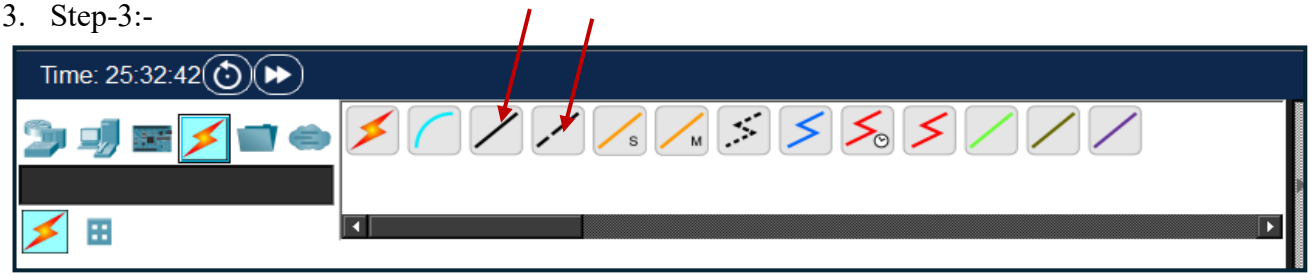
Add 10 Pc and 10 Laptop and 1 Server from this part .

2. Step-2:-



Add 5 PT-Hub and 1 Repeater from this part

3. Step-3:-



From this part we select cable to connect the component . In this lab we use 2 types of Cable. One is Copper Cross-Over Cable's and another is Normal Copper Cable's.

For Hub and laptop or pc connection And Hub to Server connection we use **Copper Cables** . Like this:-

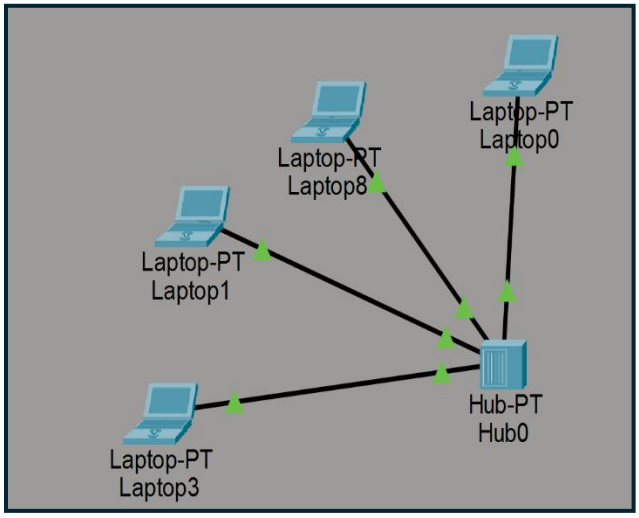


Fig:1 :- Hub to Laptop connection.

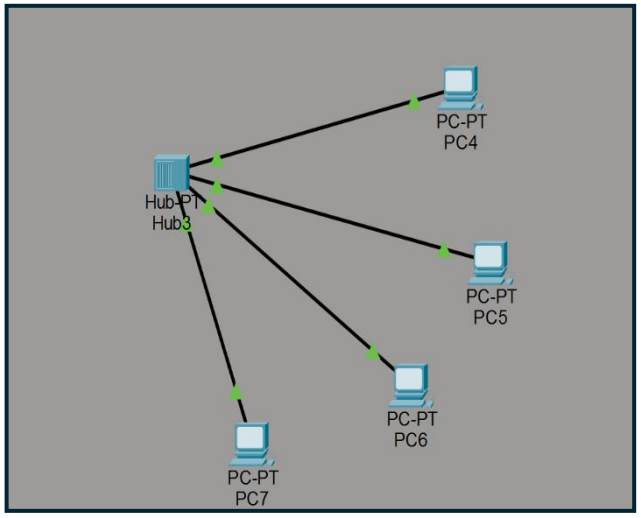


Fig:2- Hub to Pc connection.

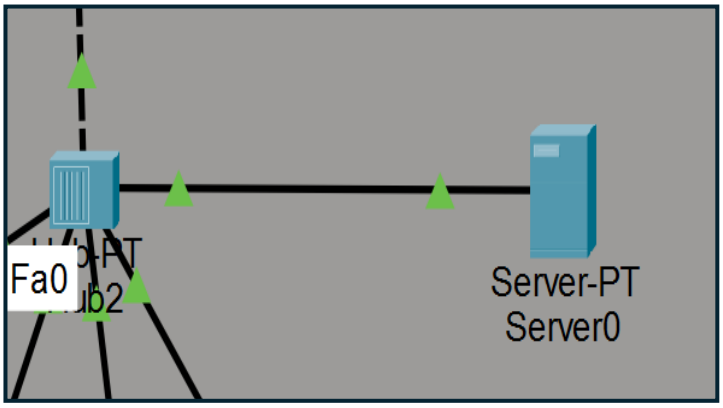


Fig:3:-Hub to Server connection.

For Hub to Hub and Repeater to Hub Connection We use **Copper Cross-Over Cable's.** Like this:-

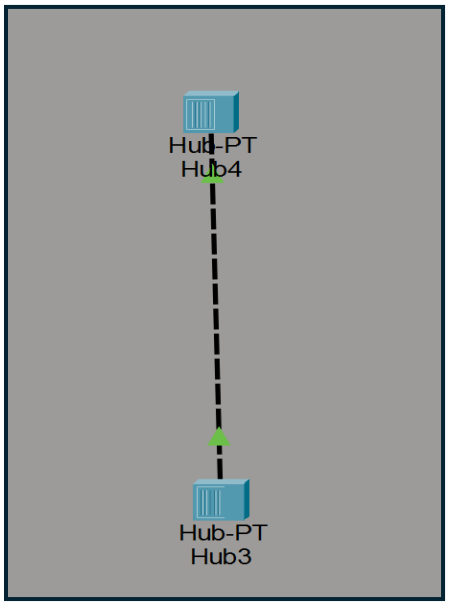


Fig:4:- Hub to Hub Connection.

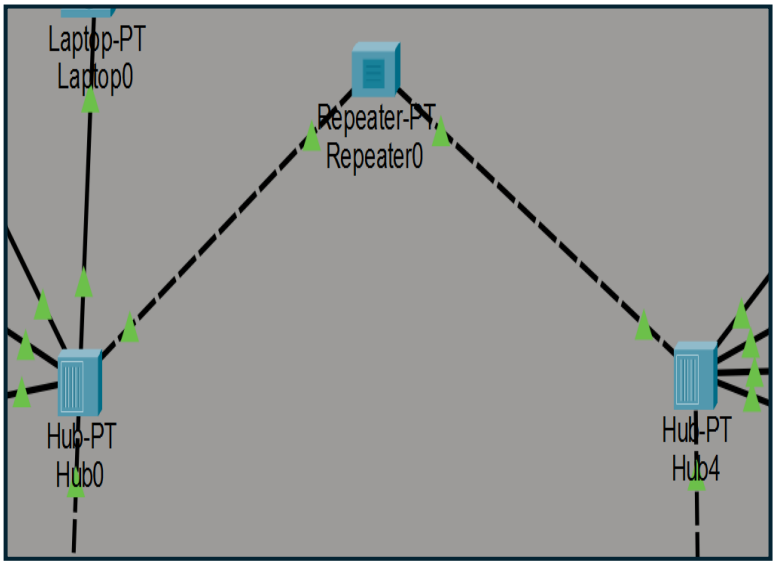
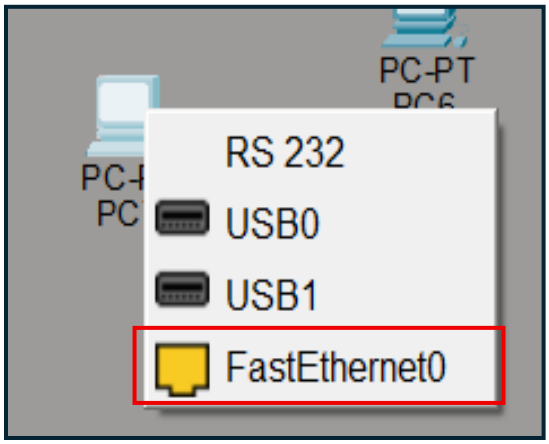
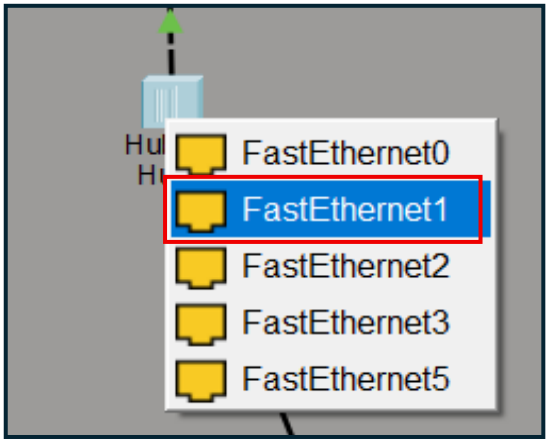


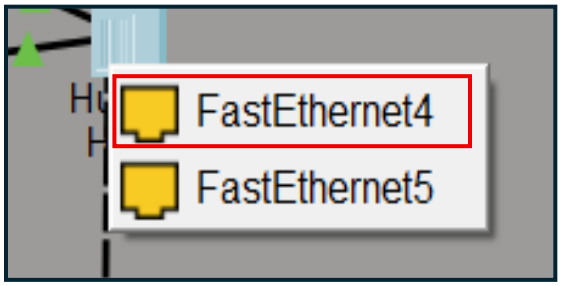
Fig:5:- Hub to Repeater Connection.

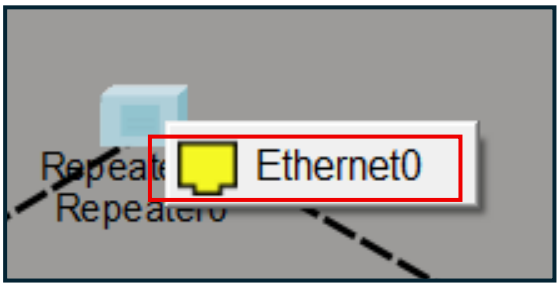
For Hub To PC or laptop Connection We use Ethernet Port:-





For Hub to Hub and Hub to Repeater Connection We also use Ethernet Port :-





After Connection The Component My Final Design is :

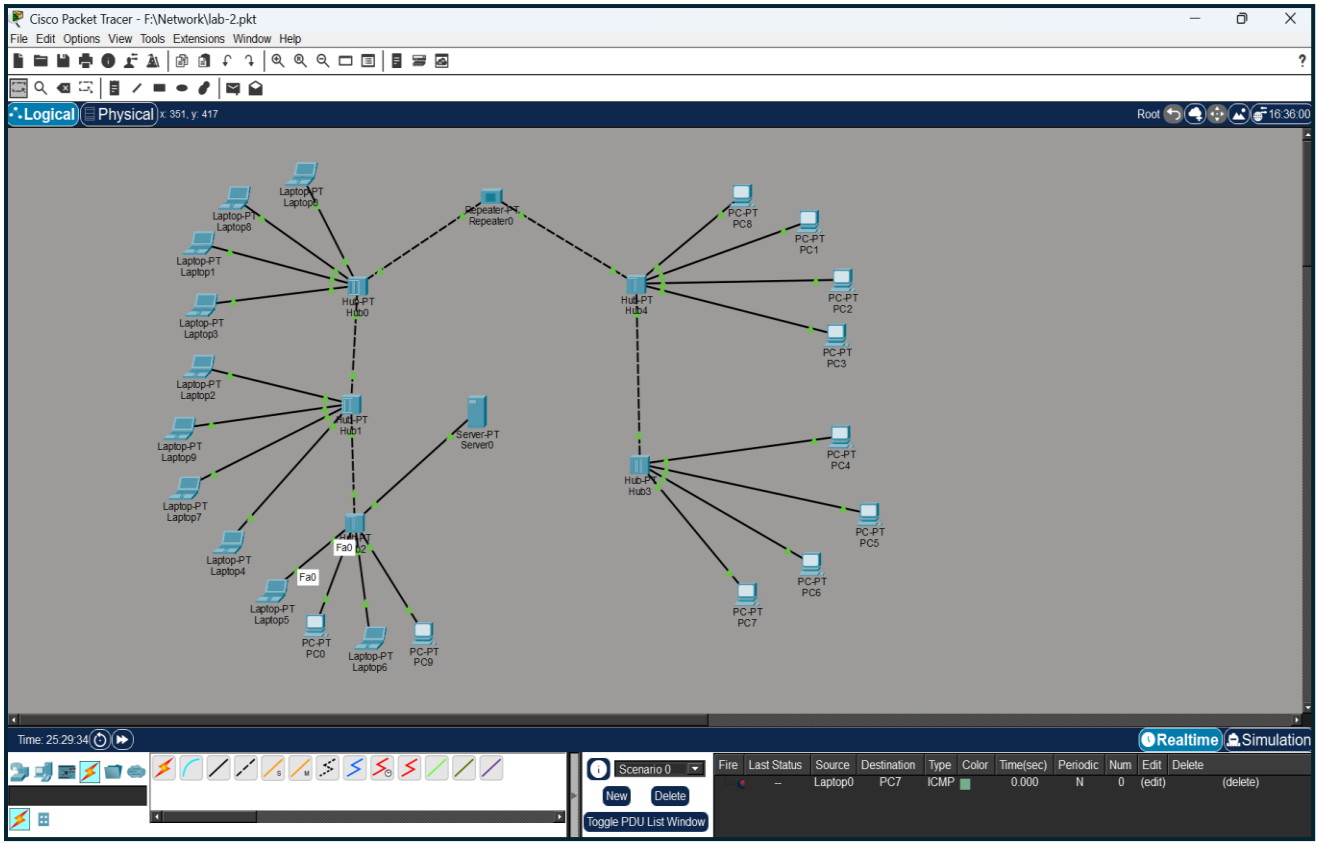
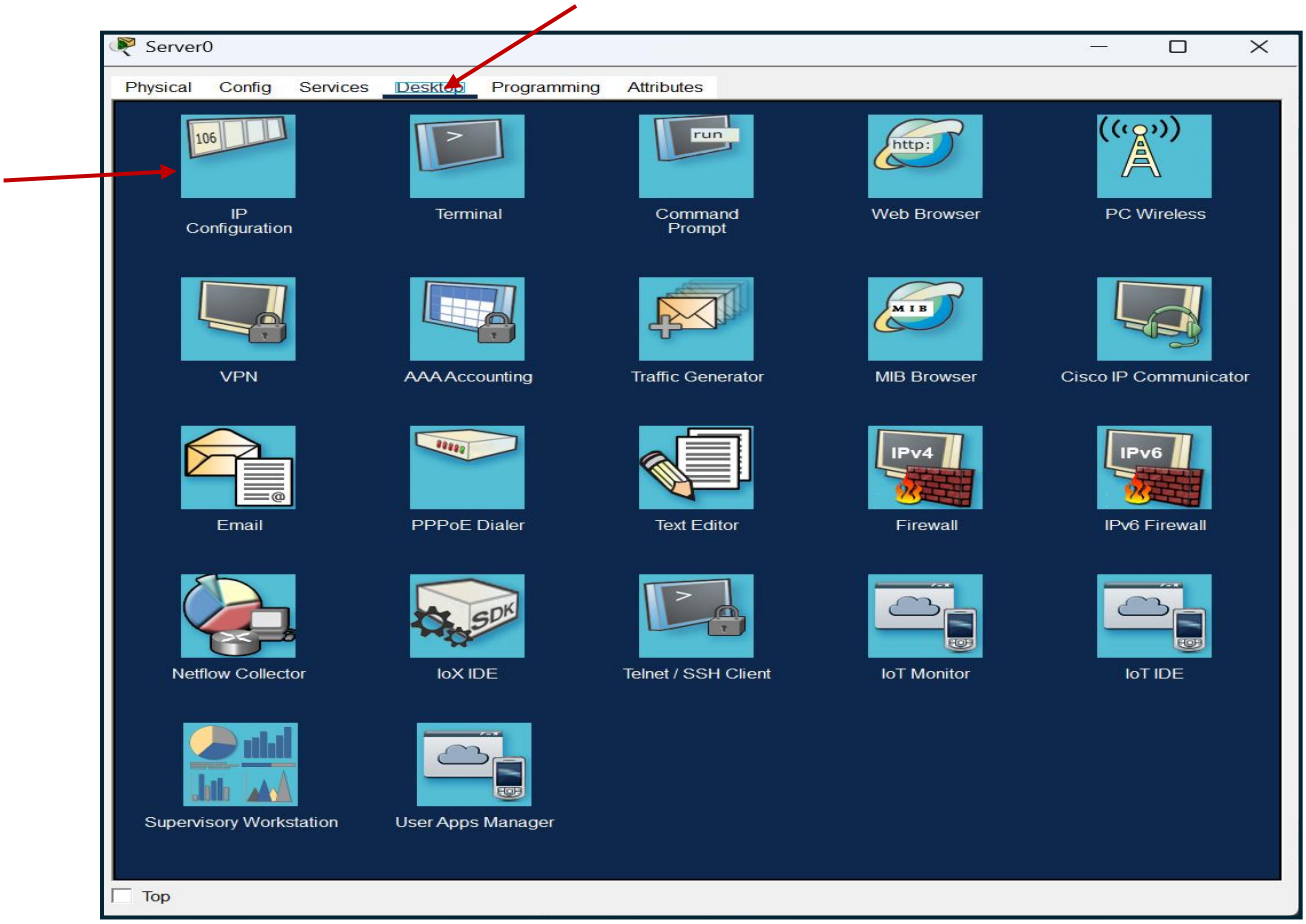


Fig:6:- Final Design.

4. Step:4:-

Now, We setup Server

Click Server → Desktop → IP Configuration

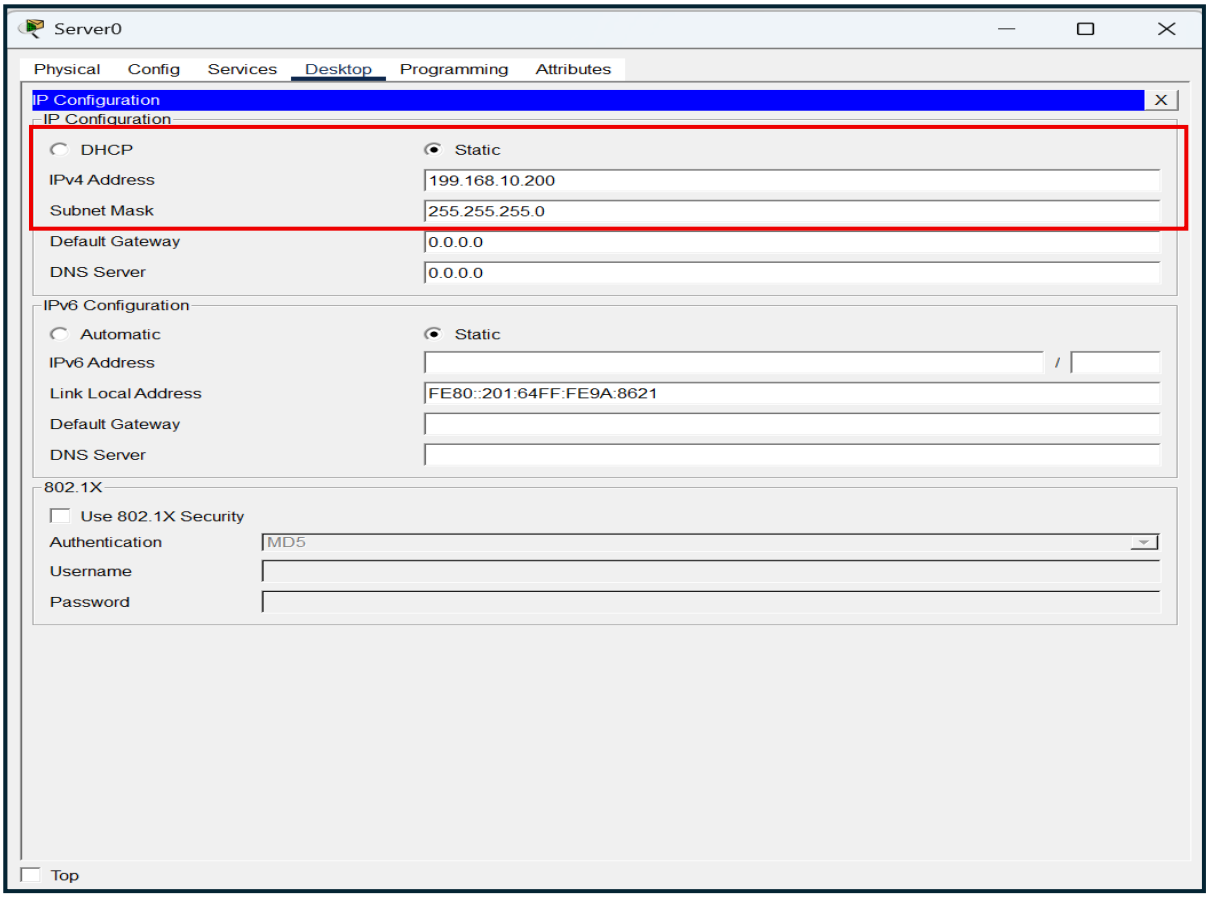


Then,

Select IP Address Static

IPv4 Address:- 199.168.10.200

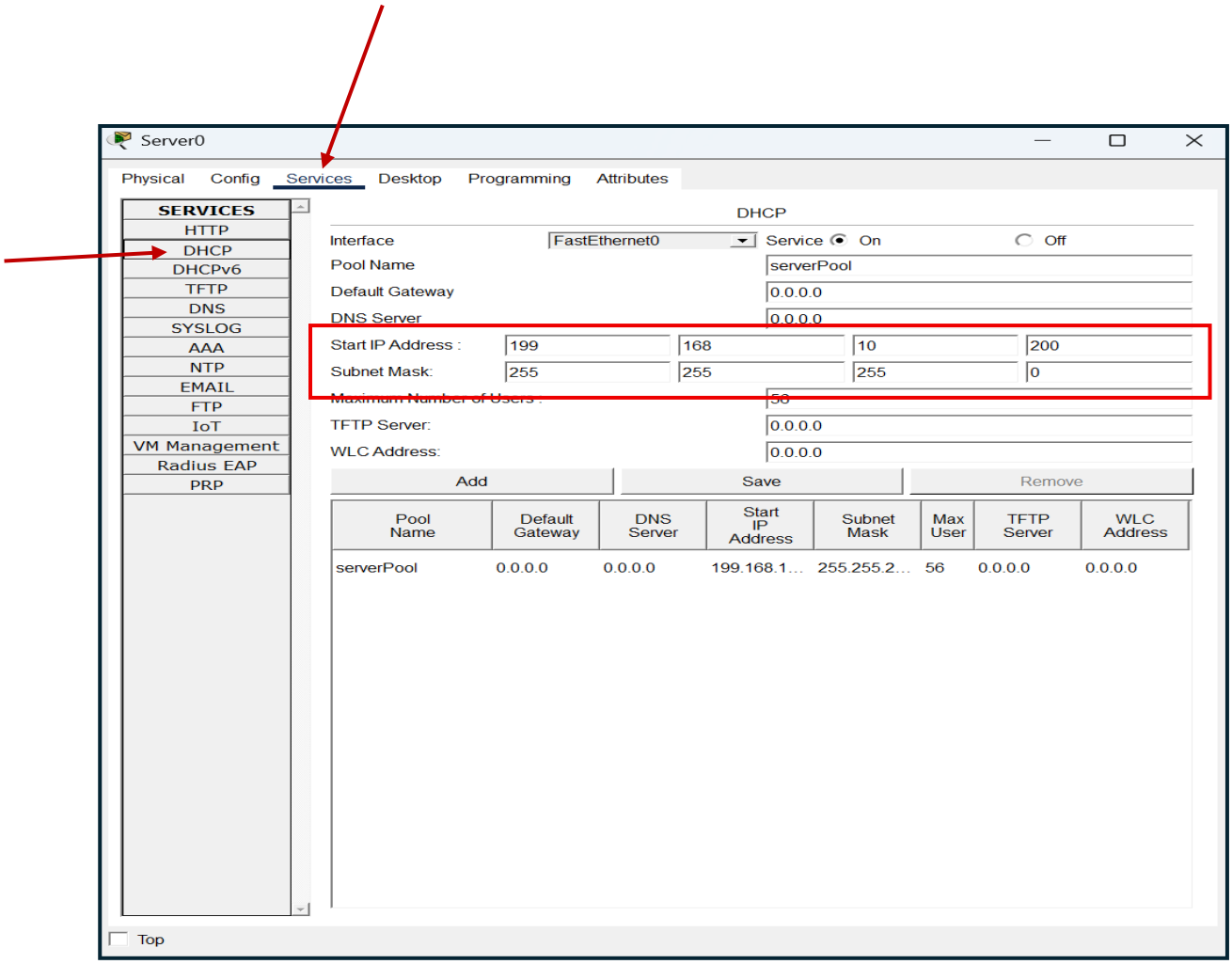
Subnet Mask :255.255.255.0



Then Go to Service → DHCP

- DHCP Service On
- Start IP Address:199.168.10.200
- Subnet Mask :255.255.255.0

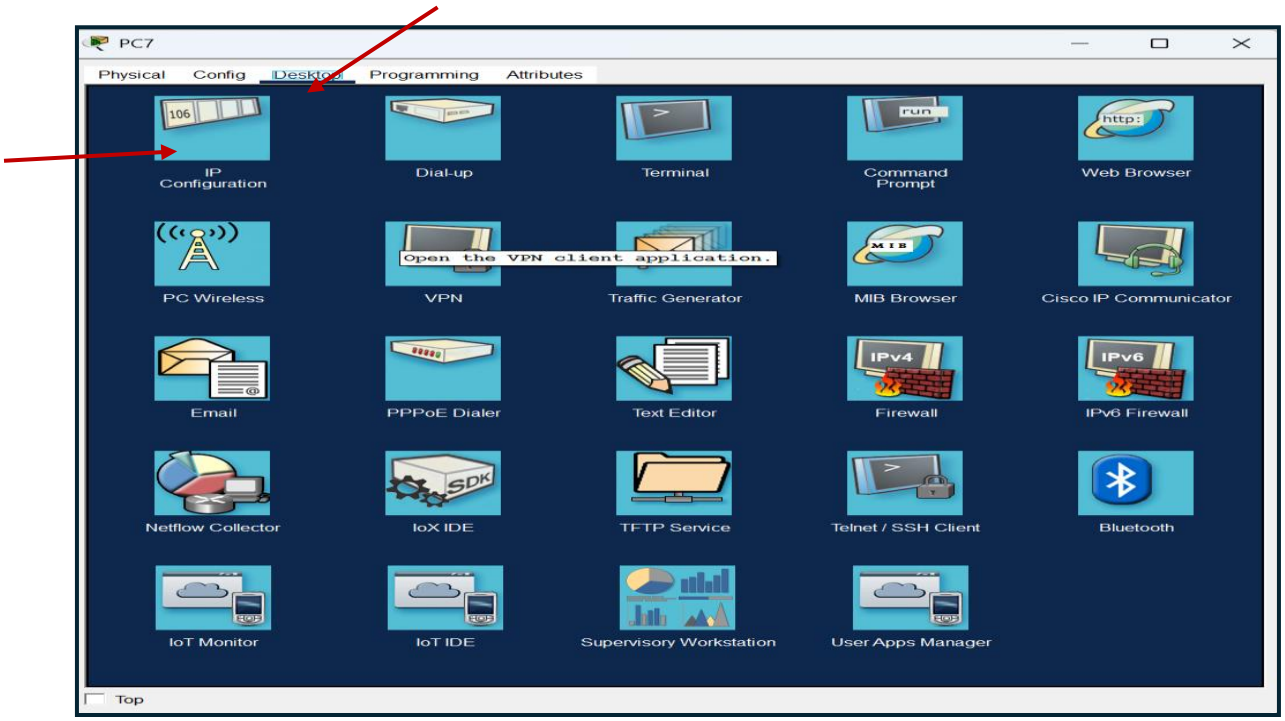
Then Save the DHCP Sitting.


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
Start IP Address :
199
168
10
200
Subnet Mask:
255
255
255
0
Maximum Number of Users :
50
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
0.0.0.0
0.0.0.0
199.168.1...
255.255.2...
56
0.0.0.0
0.0.0.0
Top

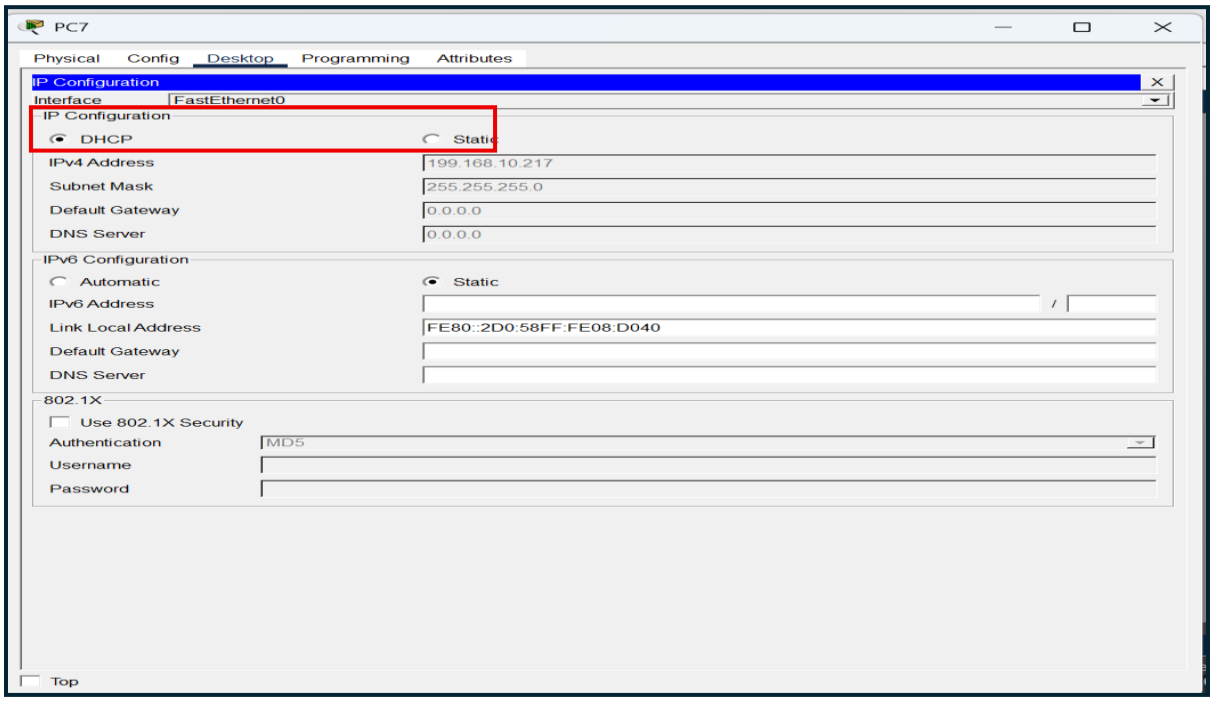
5. Step:5:-

Now , we setup All laptops and Pc Like this way:-

Click PC → Desktop → IP Configuration ->Static->IPv4 (as like picture below)



Then , Select Ip Configuration DHCP.



Since all PCs and laptops use **DHCP**, the server will assign IP addresses sequentially starting from **199.168.10.200**.

Device Name	**IP Address**	**Subnet Mask**
PC0	199.168.10.200	255.255.255.0
PC1	199.168.10.201	255.255.255.0
PC2	199.168.10.202	255.255.255.0
PC3	199.168.10.203	255.255.255.0
PC4	199.168.10.204	255.255.255.0
PC5	199.168.10.205	255.255.255.0
PC6	199.168.10.206	255.255.255.0
PC7	199.168.10.207	255.255.255.0
PC8	199.168.10.208	255.255.255.0
PC9	199.168.10.209	255.255.255.0
Laptop0	199.168.10.210	255.255.255.0
Laptop1	199.168.10.211	255.255.255.0
Laptop2	199.168.10.212	255.255.255.0
Laptop3	199.168.10.213	255.255.255.0
Laptop4	199.168.10.214	255.255.255.0
Laptop5	199.168.10.215	255.255.255.0
Laptop6	199.168.10.216	255.255.255.0
Laptop7	199.168.10.217	255.255.255.0
Laptop8	199.168.10.218	255.255.255.0
Laptop9	199.168.10.219	255.255.255.0

6. Step 6:
   Add a Simple PDU (Packet) 1.Select Add Simple PDU (envelope icon).



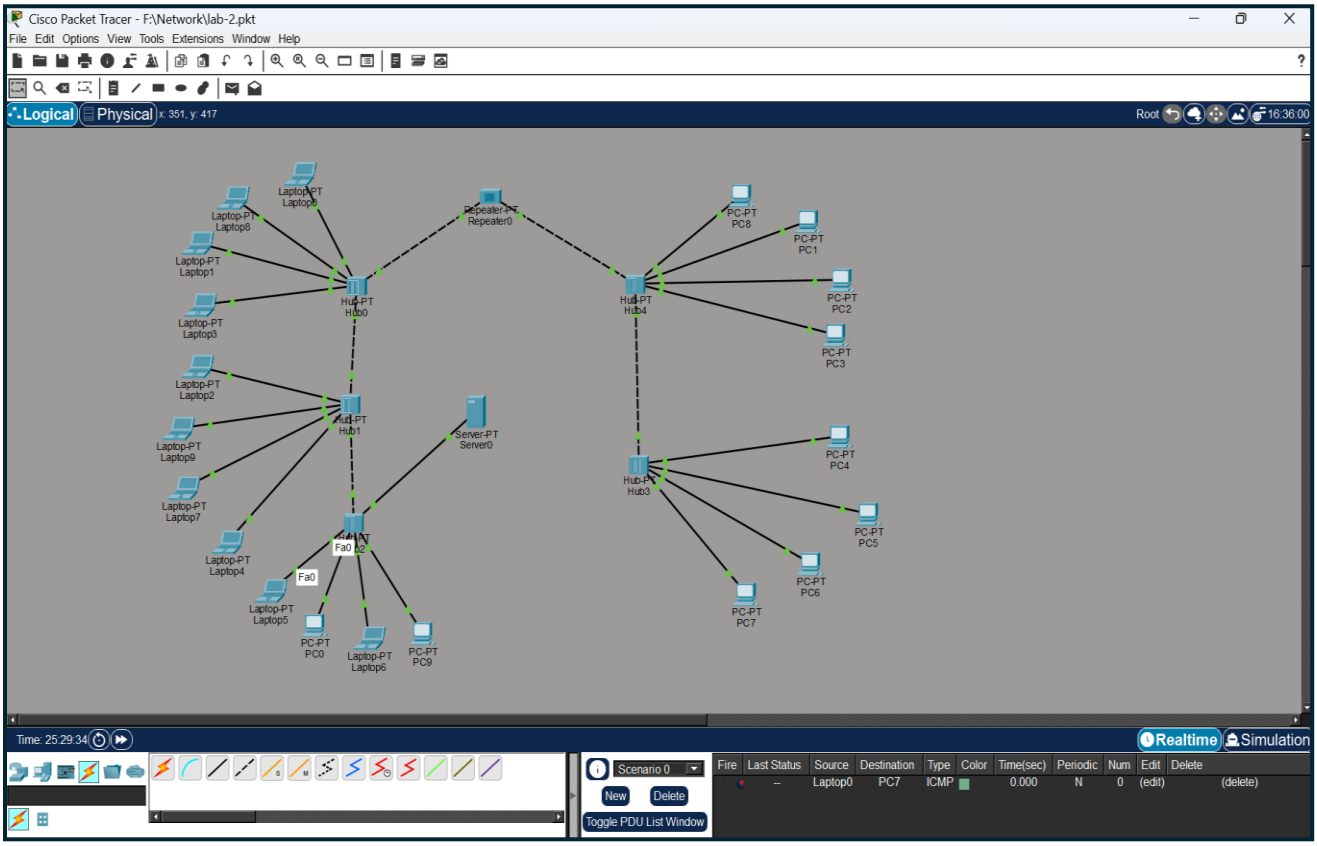
Click on Laptop 0, then on PC-6 as For Packet Transfer . Observe the envelope (packet) moving through: 10. Laptop 0→ Hub0 →Repeater 0→ Hub 4→ Hub 3 → PC-6

- The Event List showed:

. Successful → if the packet reaches PC-6 correctly.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Laptop0	PC6	ICMP		0.000	N	0	(edit)	(delete)

Final picture of completed lab task2:-



• LAB-3

Multiple devices (PCs and a laptop) across two different subnets, connected through two routers and a wireless router serving wireless clients.

Network Components

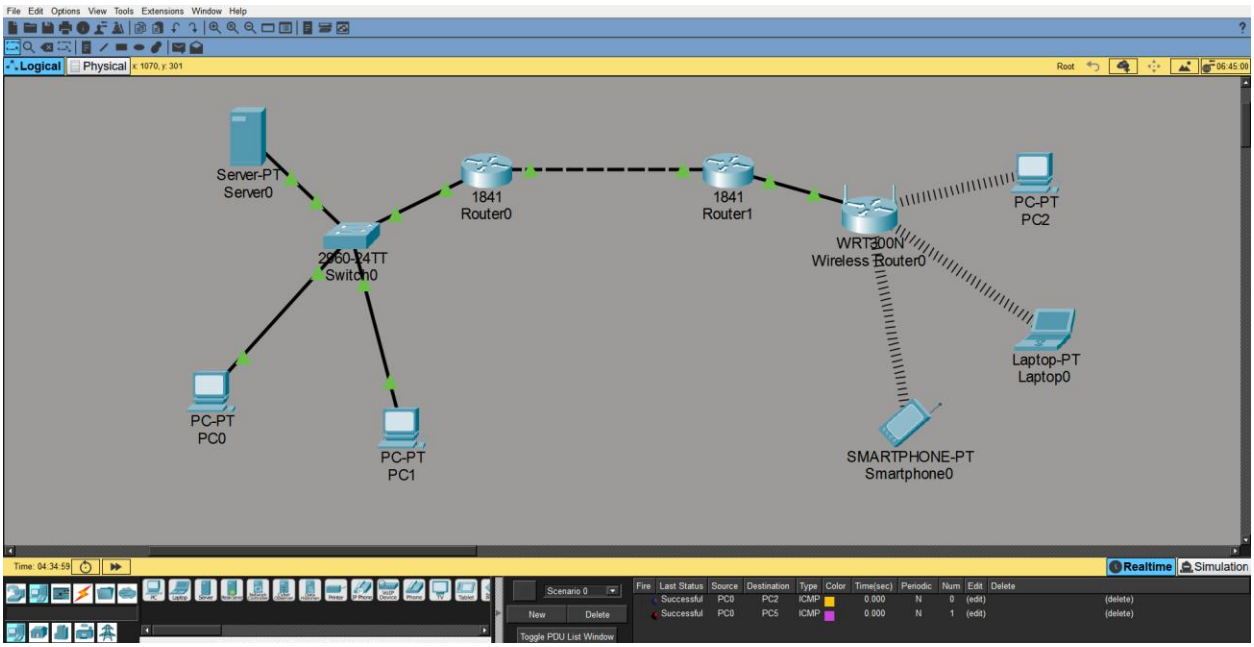
- 1 Server (Server0)
- 1 Switch (Switch0 – 2960-24TT)
- 2 Routers (Router0, Router1 – Cisco 1841)
- 1 Wireless Router (Wireless Router0 – WRT300N)
- 2 PCs on LAN (PC0, PC1)
- 1 Wireless PC (PC2)
- 1 Wireless Laptop (Laptop0)
- 1 Wireless Smartphone (Smartphone0)

Network Topology Description

- Switch0 connects Server0, PC0, and PC1 to Router0 (LAN segment).
- Router0 is connected to Router1 using a Copper Cross-Over cable (Router-to-Router serial/WAN link).
- Router1 is connected to Wireless Router0 using a Copper Straight-Through cable.
- Wireless Router0 wirelessly connects PC2, Laptop0, and Smartphone0.

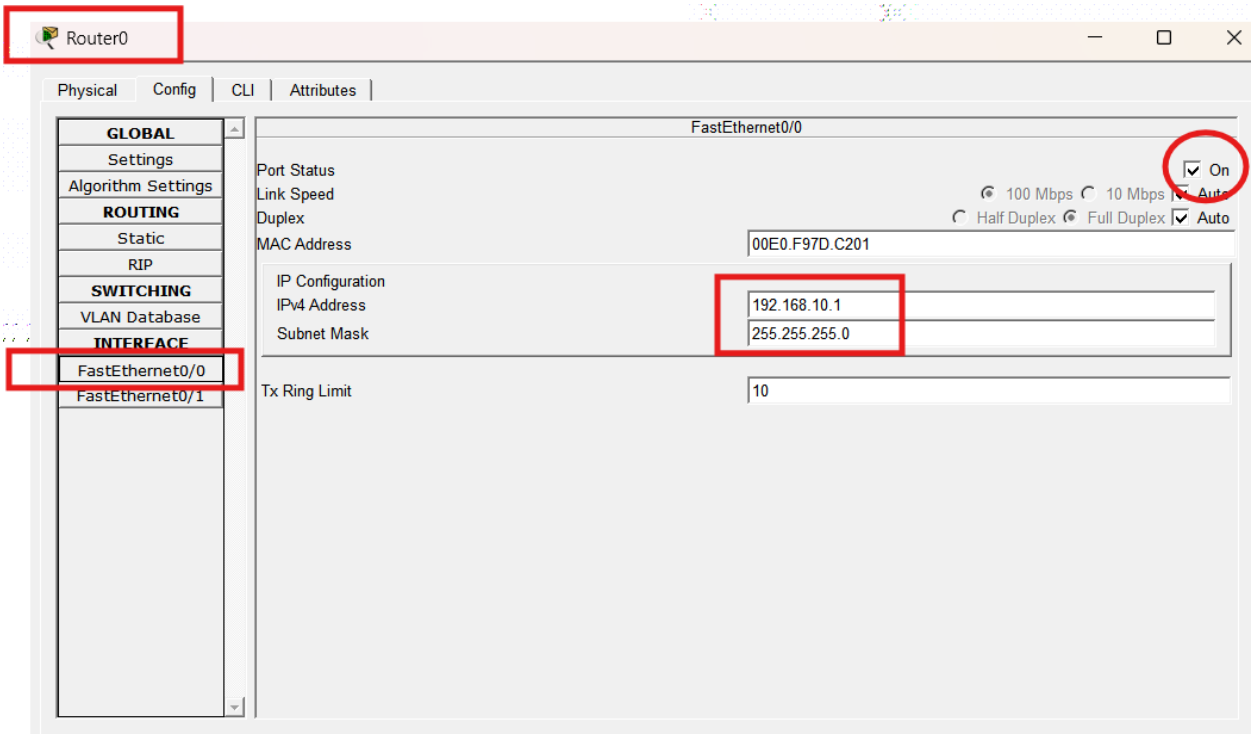
Step 1: Add Devices and Connect as like lab 1 & 2

1. Place Server0, Switch0, Router0, Router1, Wireless Router0, PC0, PC1, PC2, Laptop0, and Smartphone0 on the workspace.
2. Connect Server0, PC0, and PC1 to Switch0 using Copper Straight-Through cables.
3. Connect Switch0 to Router0 using a Copper Straight-Through cable.
4. Connect Router0 to Router1 using a Copper Cross-Over cable.
5. Connect Router1 to Wireless Router0 using a Copper Straight-Through cable.
6. Leave PC2, Laptop0, and Smartphone0 unconnected physically — they will join Wireless Router0's network wirelessly.

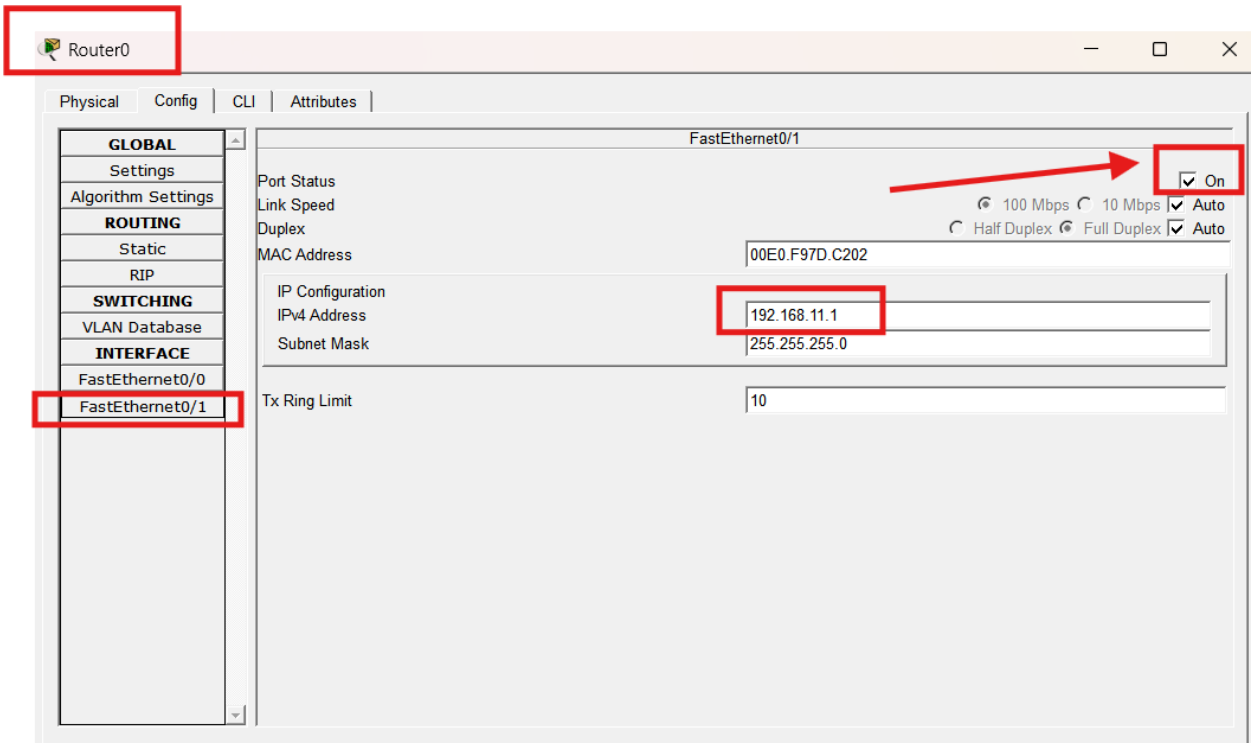


Step 2: Configure Router0 (as like Lab 3)

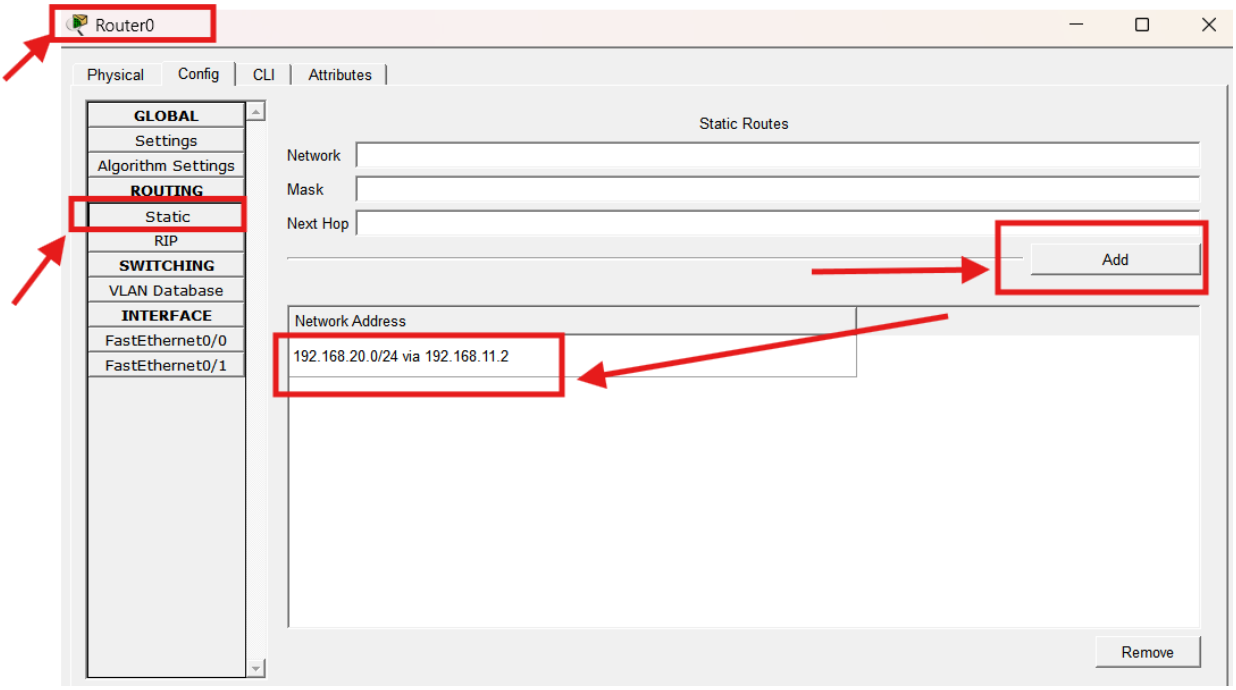
1. FastEthernet0/0 (LAN side): IP Address 192.168.10.1, Subnet Mask 255.255.255.0, Port Status On.



2. FastEthernet0/1 (WAN side, to Router1): IP Address 192.168.11.1, Subnet Mask 255.255.255.0, Port Status On.

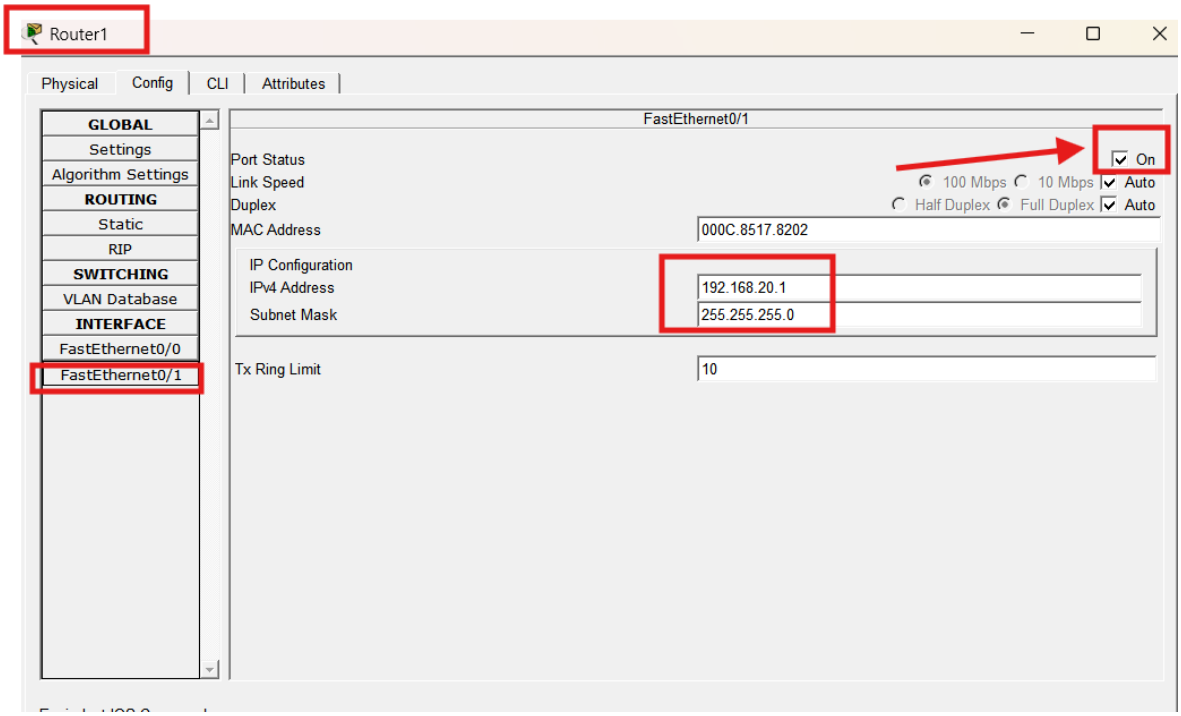


3. Configure static routes on Router0 so that traffic can pass between 192.168.10.x, 192.168.11.x and 192.168.12.x networks.

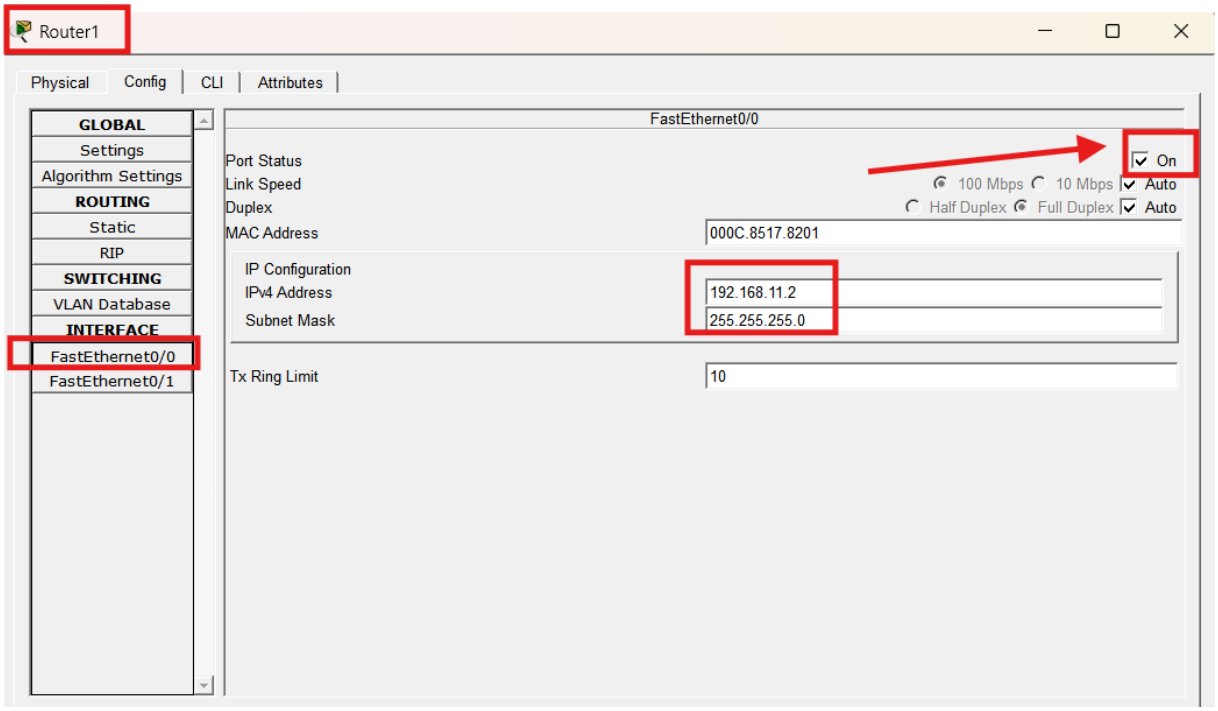


Step 3: Configure Router1

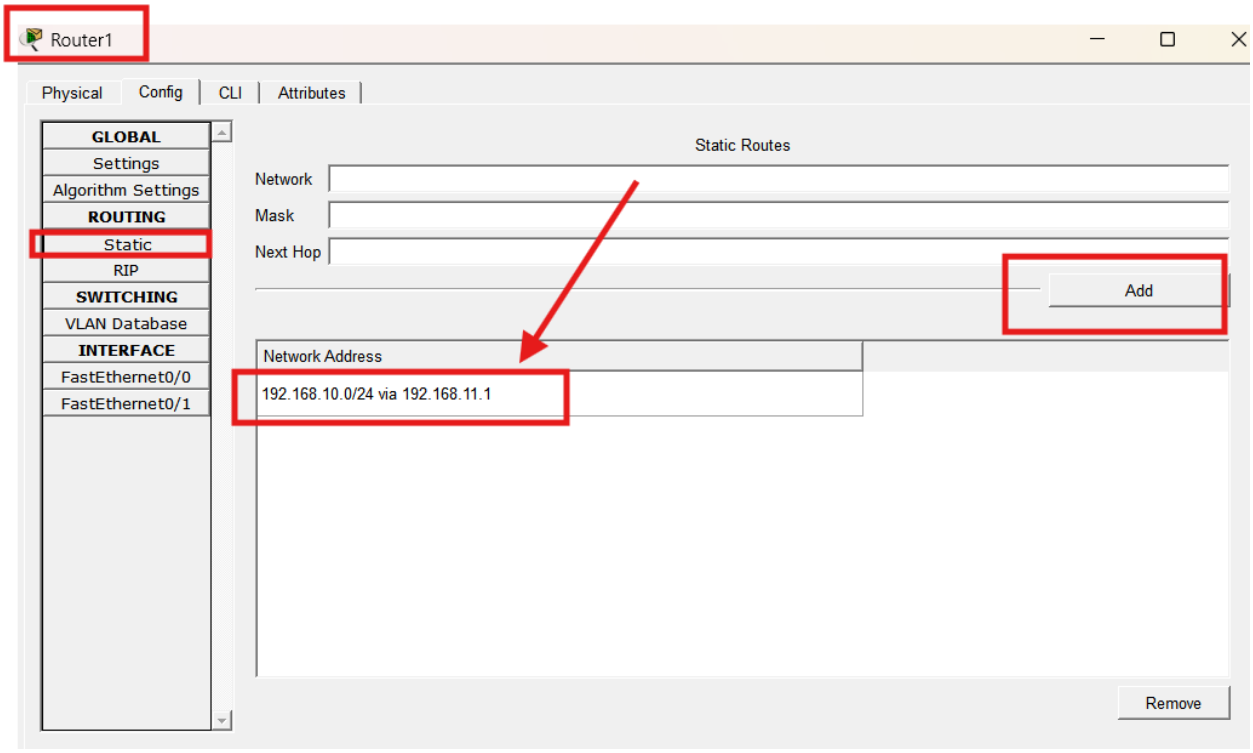
1. FastEthernet0/1 (WAN side, to Router0): IP Address 192.168.11.2, Subnet Mask 255.255.255.0, Port Status On.



2. FastEthernet0/0 (to Wireless Router0): IP Address 192.168.12.1, Subnet Mask 255.255.255.0, Port Status On.

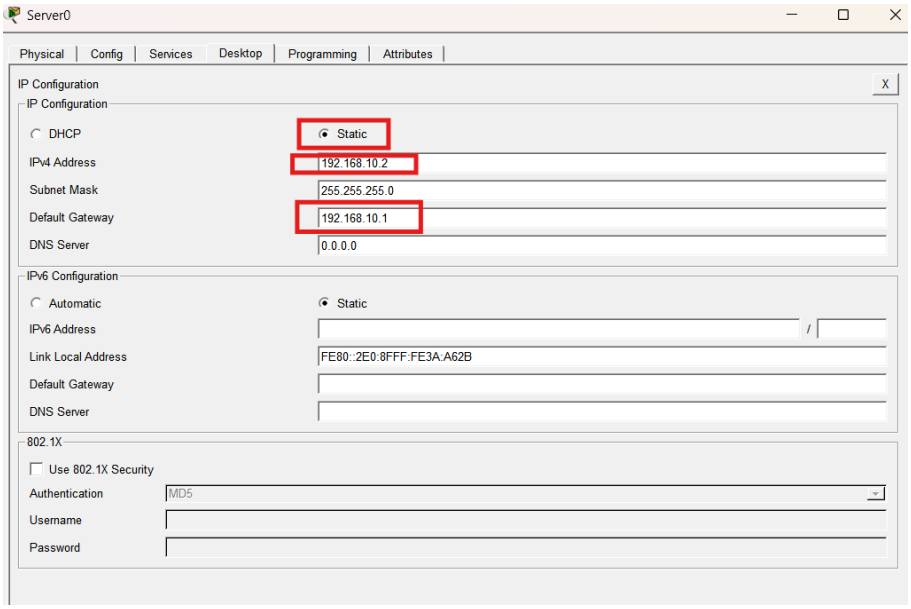


3. Configure static routes on Router1 so that traffic can pass between 192.168.10.x, 192.168.11.x and 192.168.12.x networks.

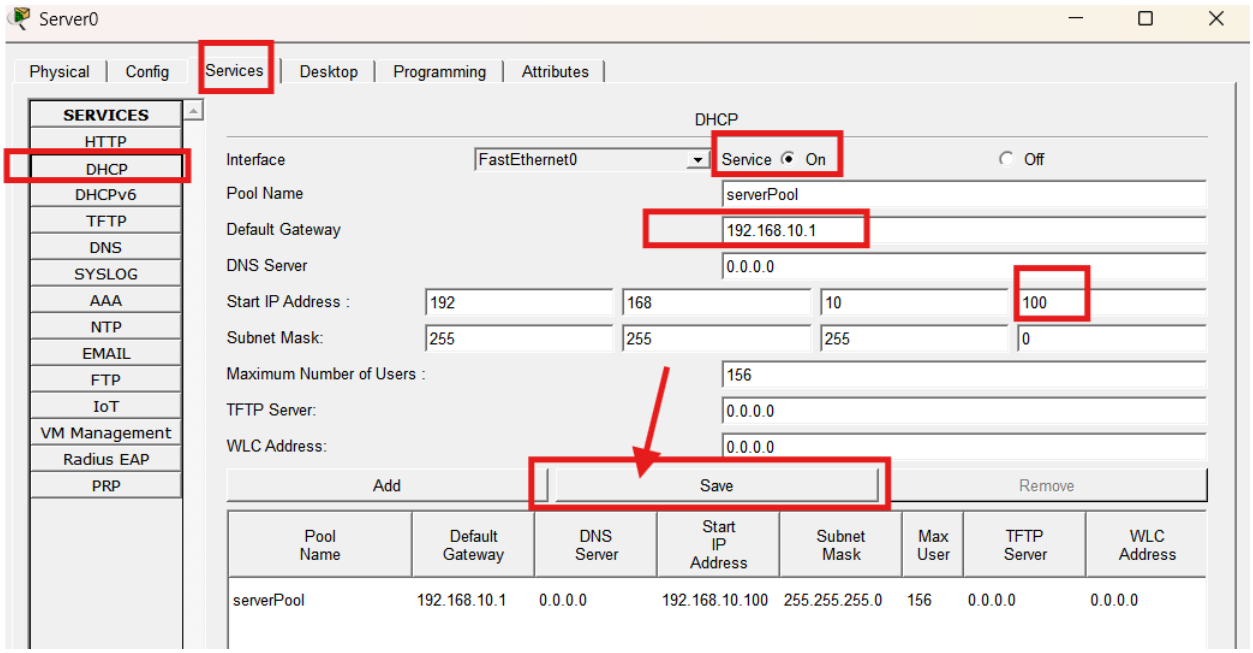


Step 4: Assign IP Addresses to LAN Devices (as like Lab 1)

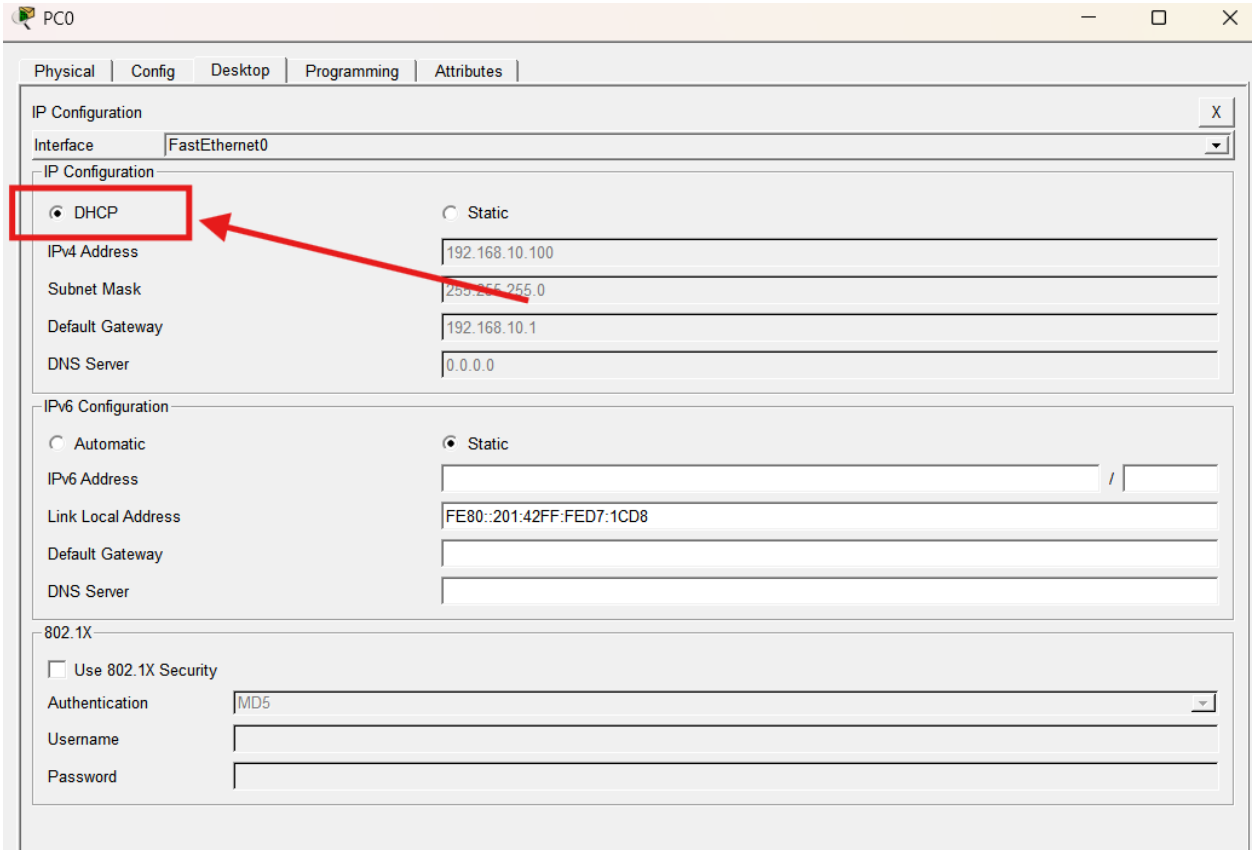
1. Server0 → Desktop → IP Configuration → Static → 192.168.10.2 / 255.255.255.0 / Gateway 192.168.10.1



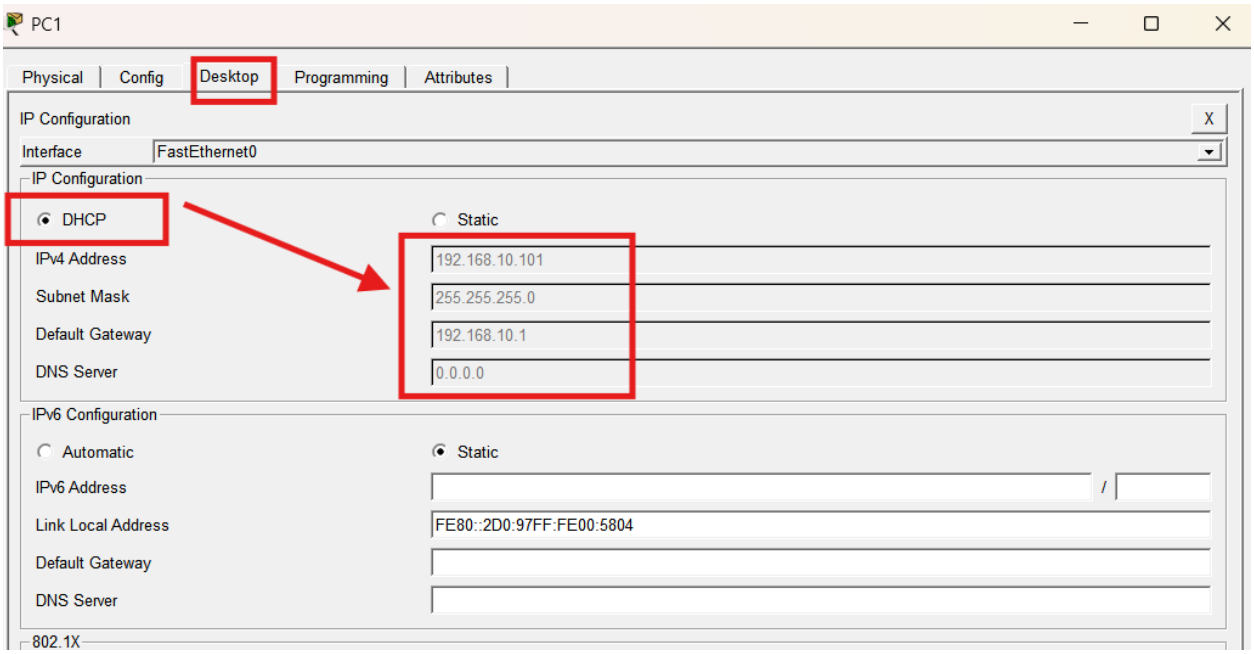
2. Server0 → Services → DHCP→ Service On → Start IP address 192.168.10.100 / 255.255.255.0 / Gateway 192.168.10.1



3. PC0 → Desktop → IP Configuration → DHCP → 192.168.10.100/ 255.255.255.0 / Gateway 192.168.10.1

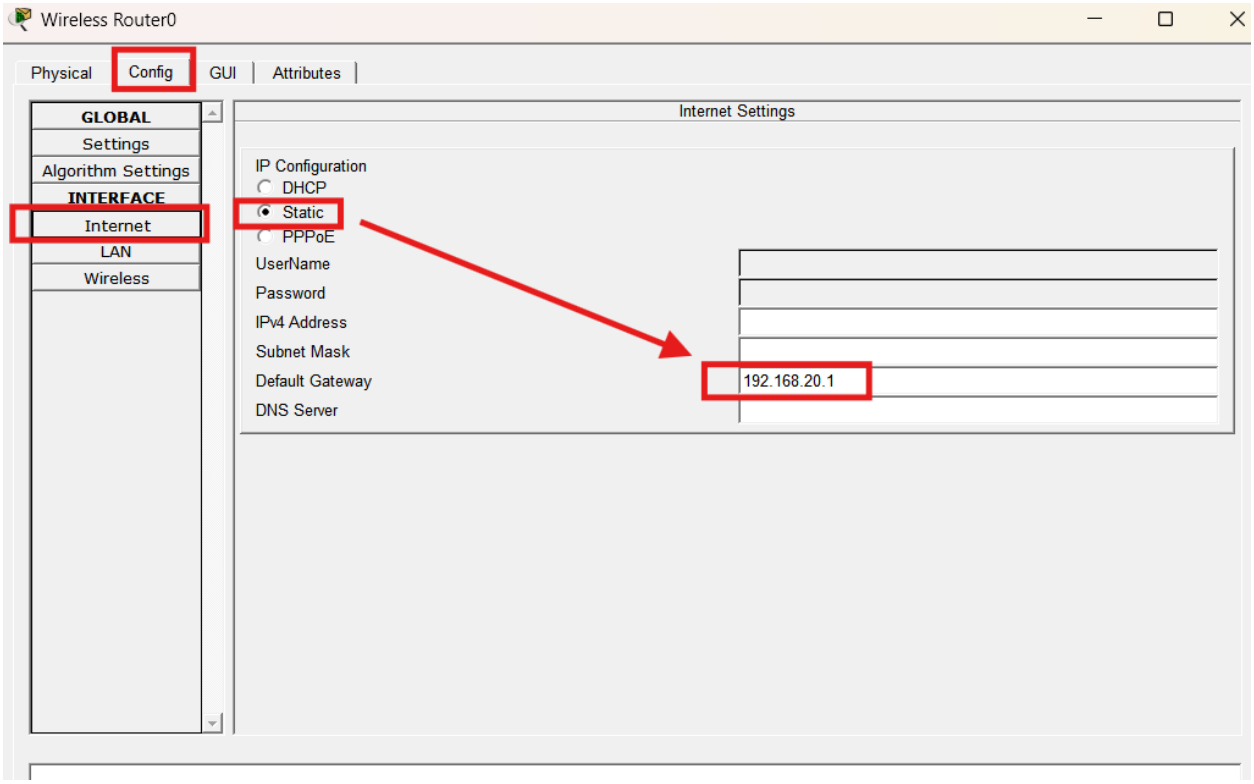


4. PC1 → Desktop → IP Configuration → DHCP → 192.168.10.101 / 255.255.255.0 / Gateway 192.168.10.1

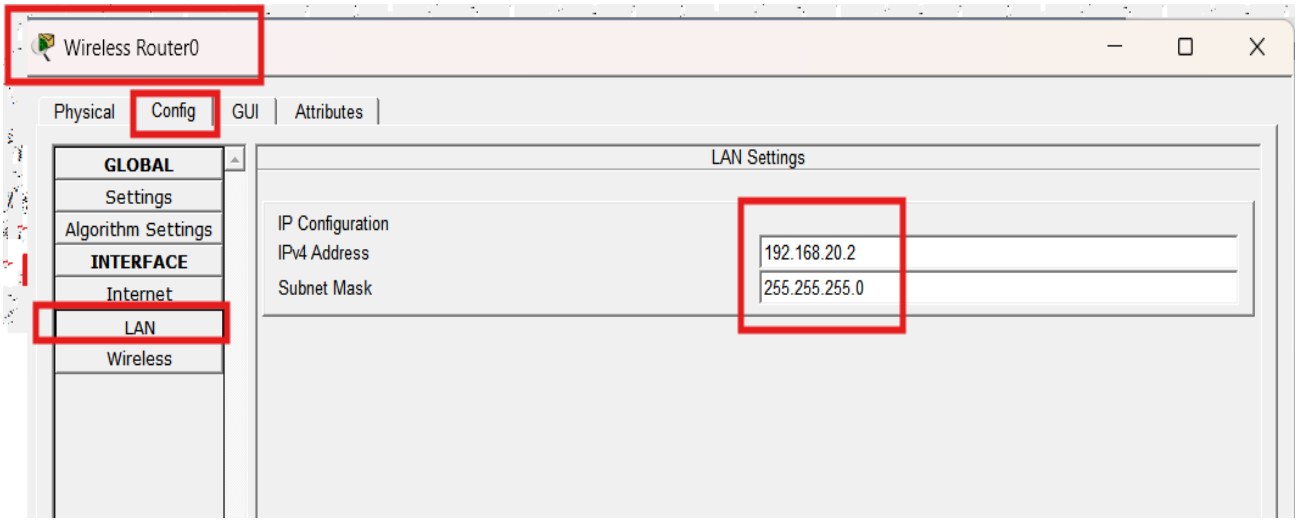


Step 5: Configure Wireless Router0

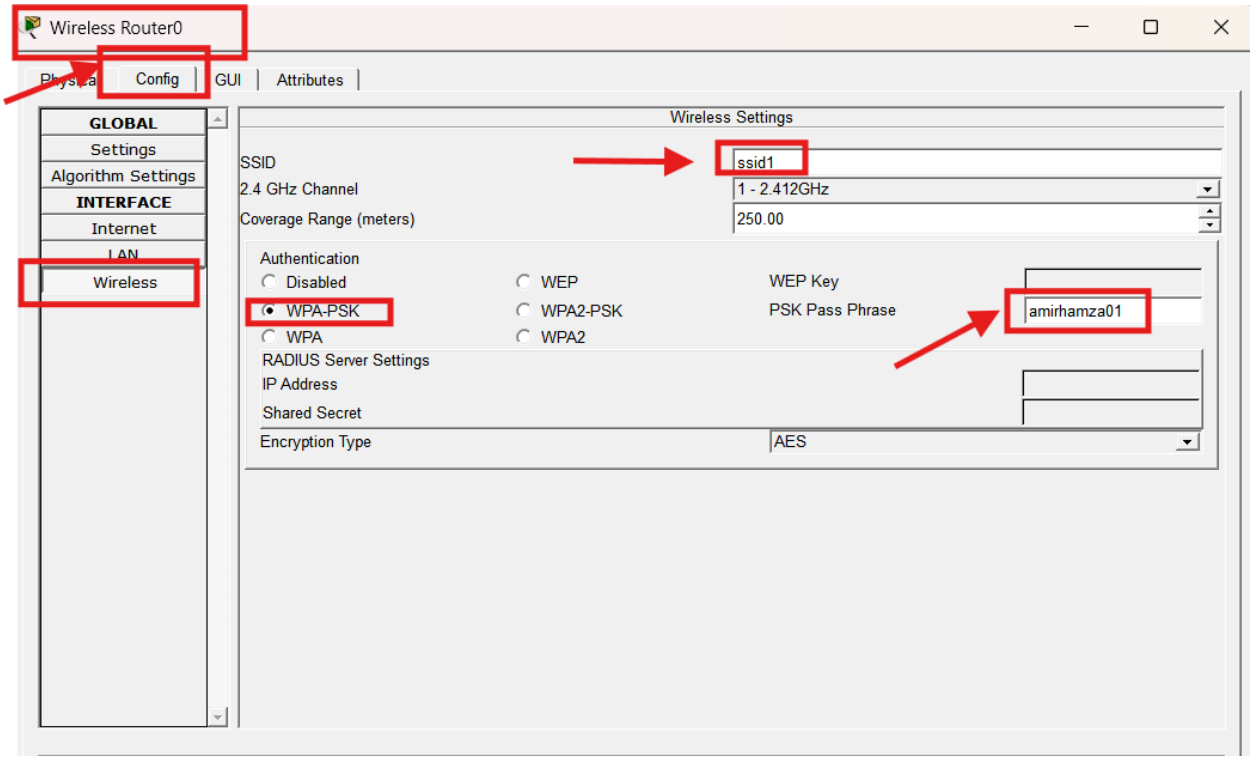
1. Go to Config → Internet: set IP Configuration to Static, Default Gateway 192.168.20.1.



2. Go to Confg → LAN Settings: Set IP Address 192.168.20.2, Subnet Mask 255.255.255.0

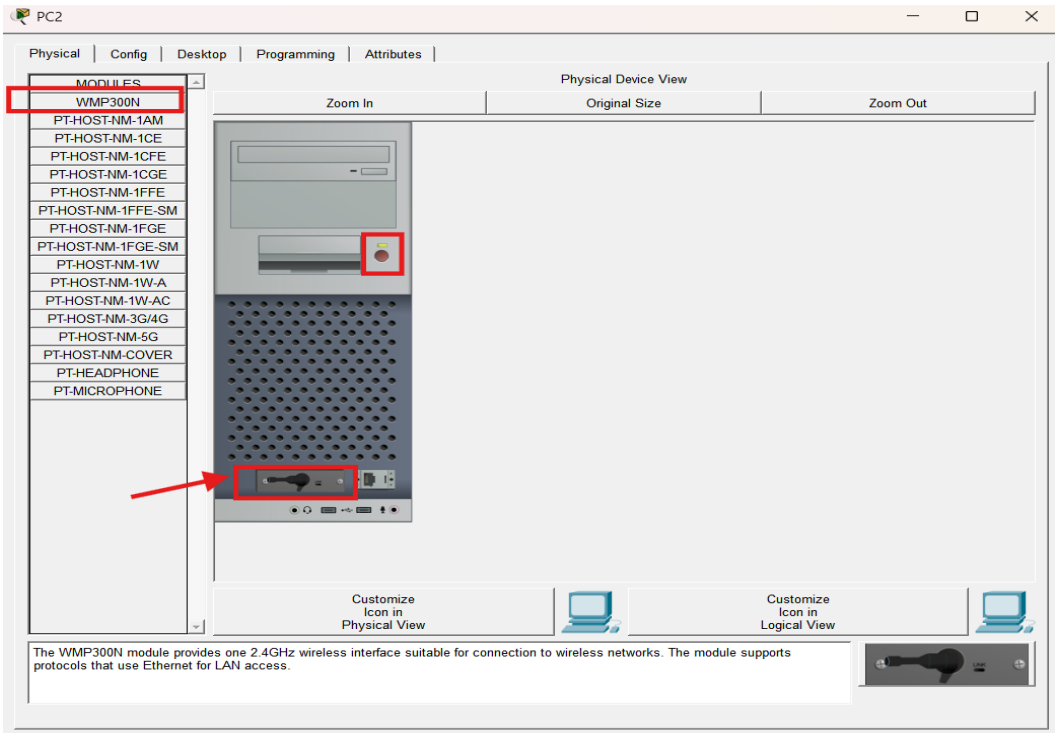


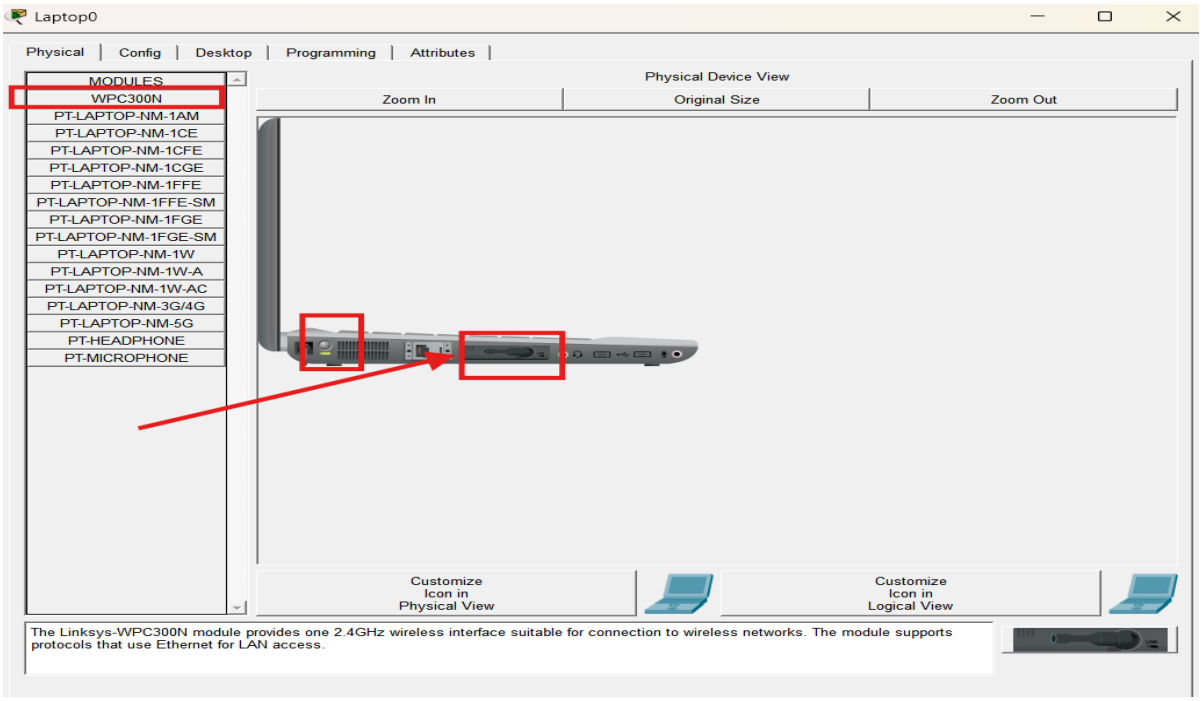
3. Go to Config → Wireless: set SSID (e.g., "ssid1"), Authentication WPA/PSK, and set a password/key [e.g., amirhamza01]



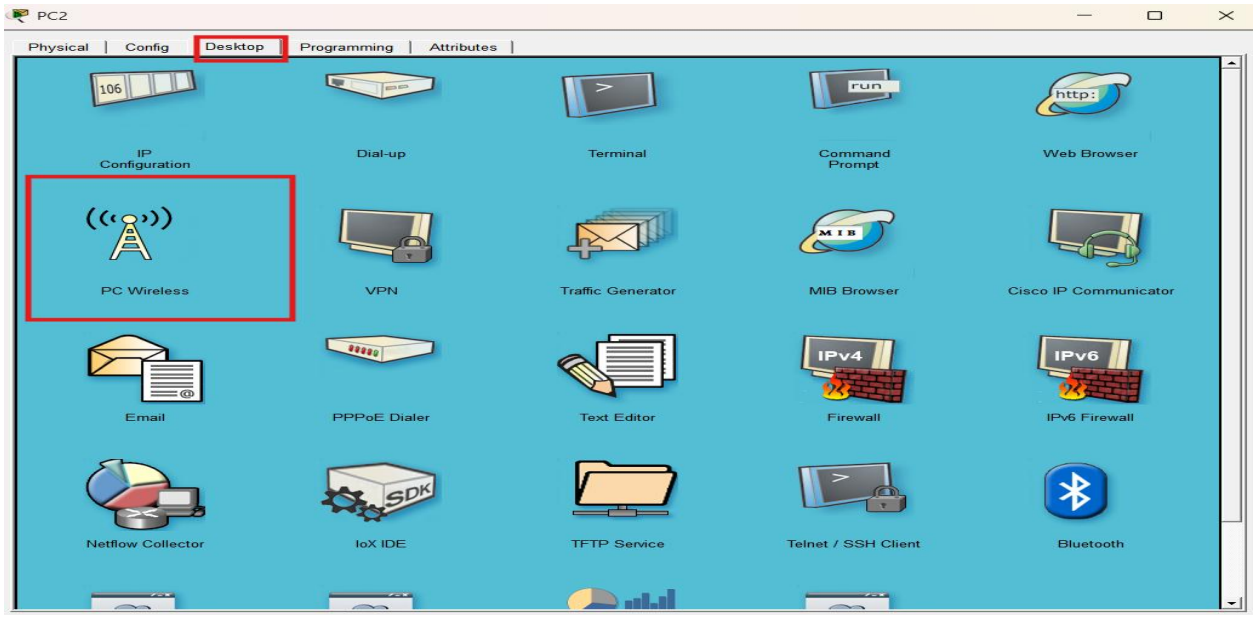
Step 6: Connect Wireless Clients

1. For PC2 and Laptop0: add a wireless (WMP300N) module — Physical tab → power off → drag module to slot → power on.



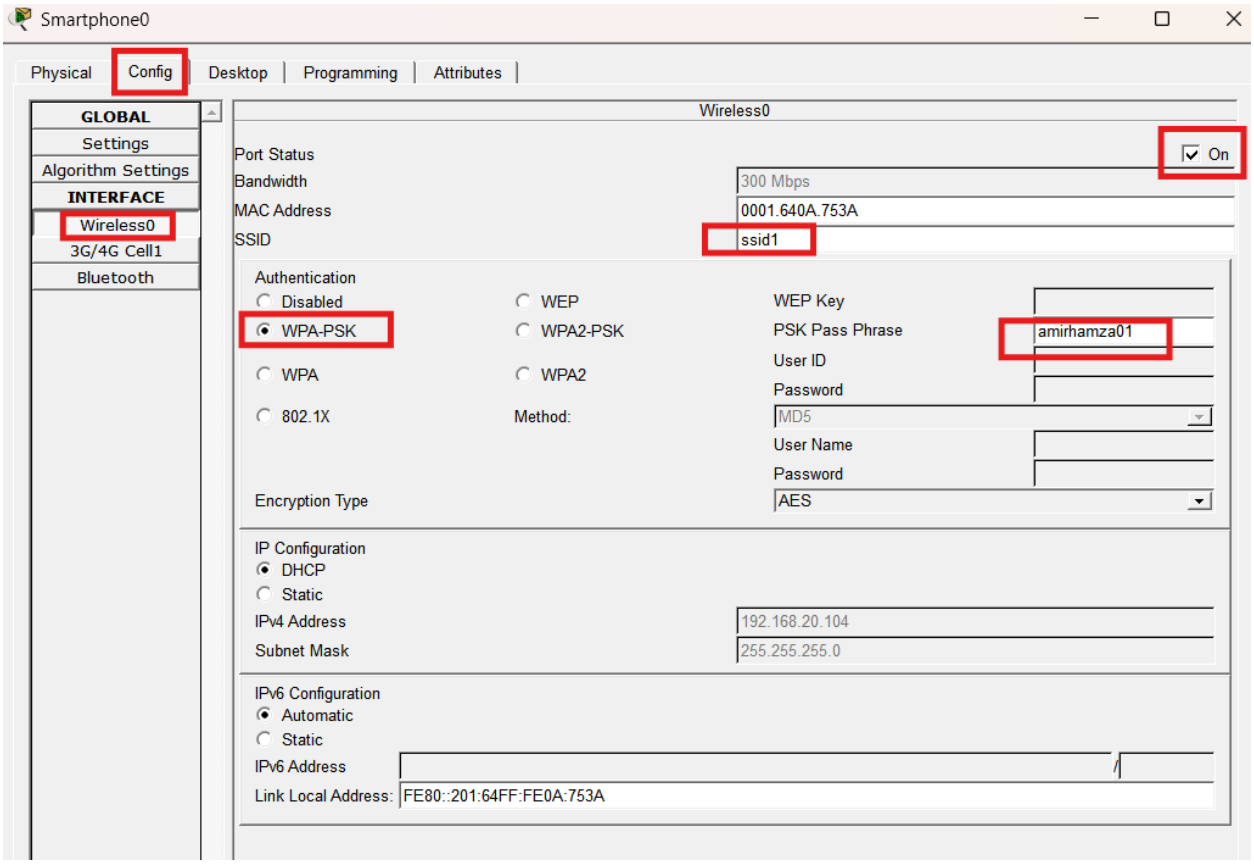


2. Go to Desktop → PC Wireless / Laptop Wireless → select the SSID → enter the password → Connect.

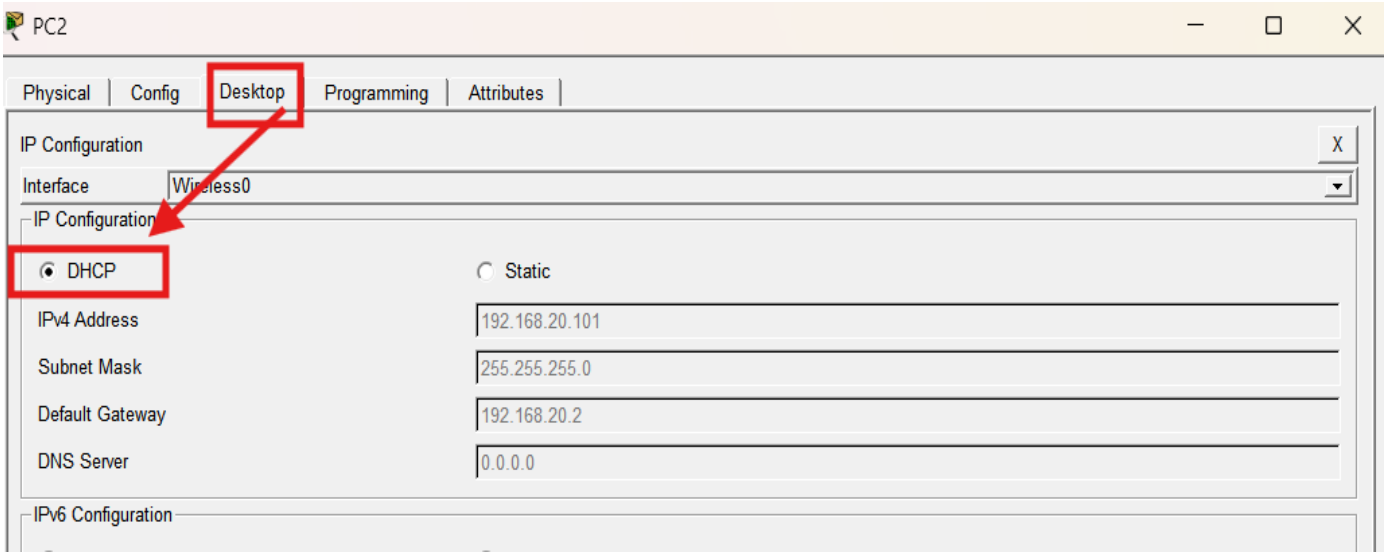




3. For Smartphone0: go to Confg → Wireless0 → set the SSID →Authentication WPA-PSK →enter the password .

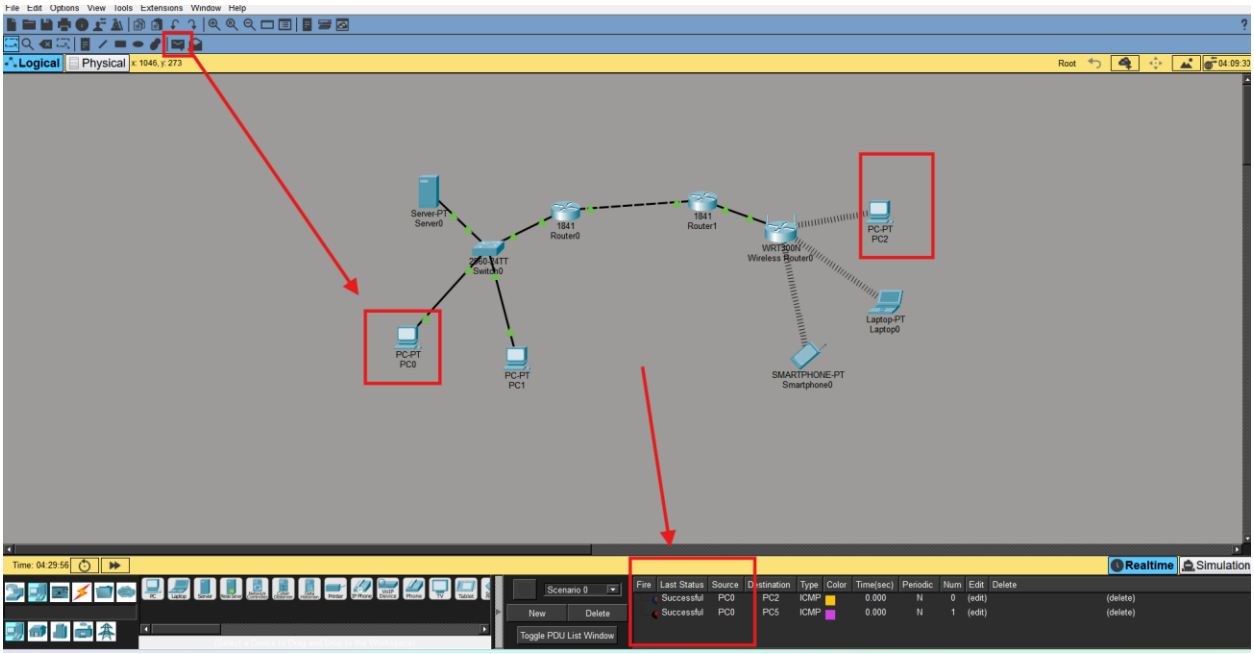


4. Set each wireless device's IP Configuration to DHCP so it automatically receives an IP in the 192.168.20.x range.



Step 7: Packet Transfer / Verify Connectivity (as like Lab 1)

1. Click (Message icon).
2. Select PC0 (LAN side) as the source and PC2 or Laptop0 (wireless side) as the destination.



The Event List should show the packet status as "Successful" if the entire path — LAN → Router0 → Router1 → Wireless Router0 → wireless client — is configured correctly.

LAB-4

Required Devices:-

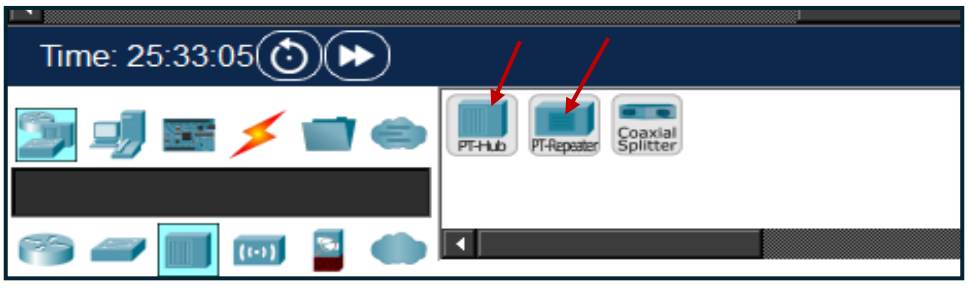
- 1 Laptop
- 1 Smartphone
- 6 Pcs
- 2 Hub-PT
- 1 Server-PT
- 1 Repeater
- 3 2960-24TT Switch
- 2 1841 Router
- 1 Wireless Router
- 1 Bridge-PT
- Copper Cross-Over Cable
- Copper Cable

1. Step-1:

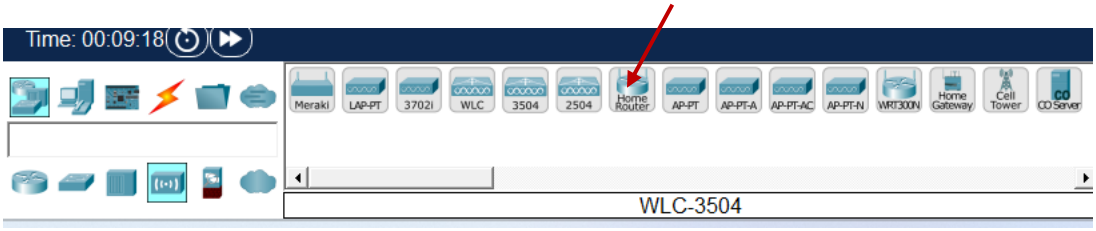


Add 6 Pc , 1 Laptop , 1 Server and 1 smart phone from this part .

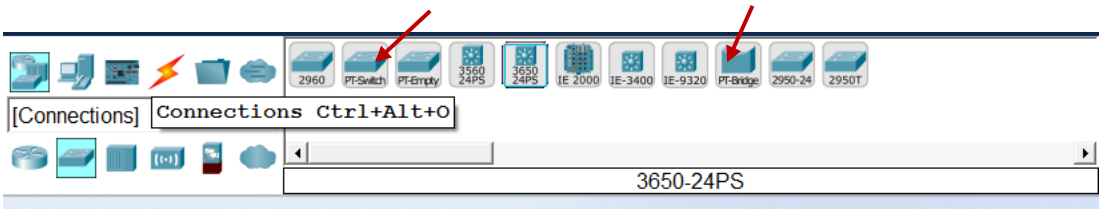
2. Step-2:-



Add 2 PT-Hub and 1 Repeater from this part



From here we add wireless Router .



From here we add Switch and Bridge .



From here we add Router .

3. Step-3:-

From this part we select cable to connect the component . In this lab we use 2 types of Cable. One is Copper Cross-Over Cable's and another is Normal Copper Cable's.



we use **Copper Cables** . For:-

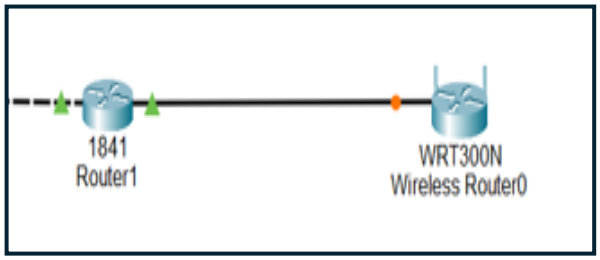


Fig:1 :- Router to wireless router connection.

.

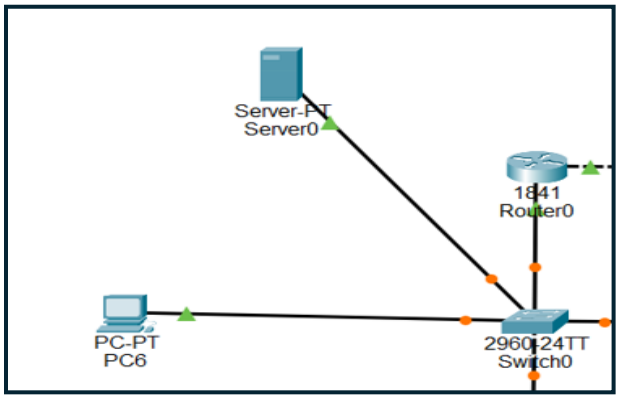


Fig:2- switch to pc , switch to Server and Switch to Router connection



Fig:3:-Switch to Switch Connection.

We use **Copper Cross-Over Cable's.** For:-

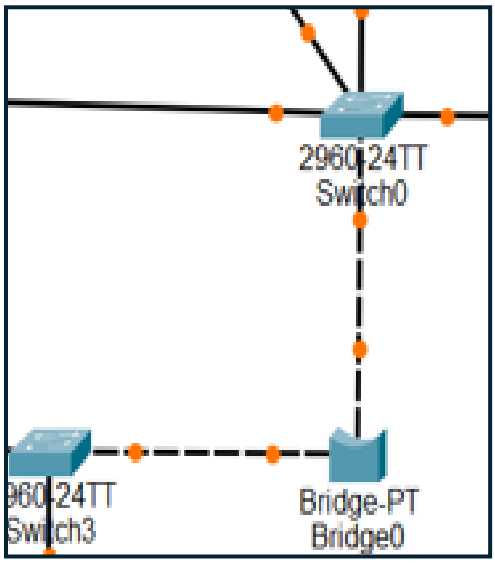


Fig:4:- Switch to Bridge Connection

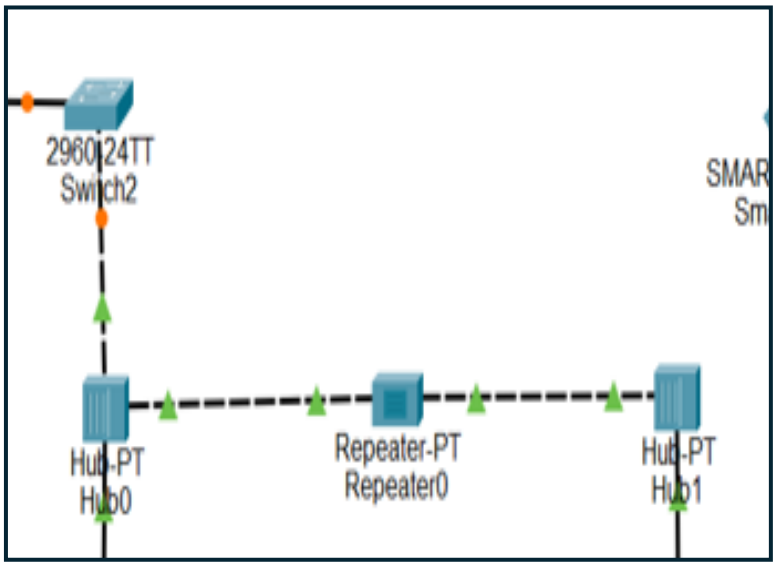


Fig:5:- Hub to Repeater Connection. And Hub to Switch Connection

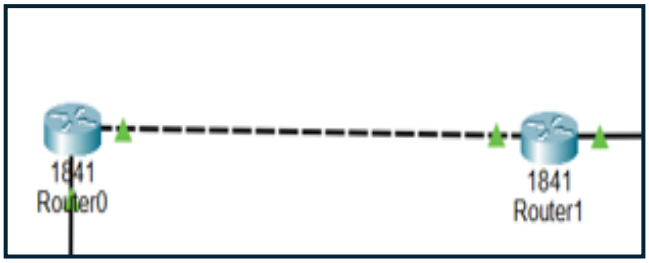
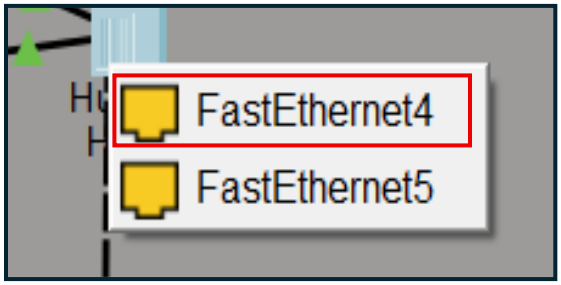
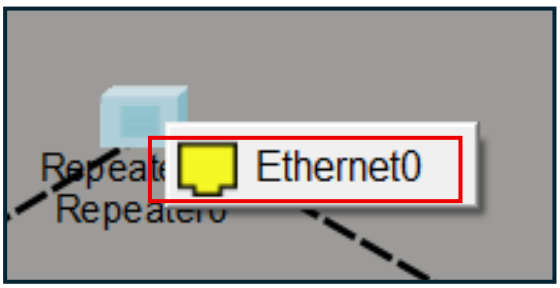


Fig :6:- Router to Router Connection

- Connect Router to Router By Ethernet Port.
- Switch to Brich By Ethernet Port .
- Weird Pc connect by Ethernet Port

For Hub to Hub and Hub to Repeater Connection We also use Ethernet Port :-





4. Step:4:-
   Setup Server Like (Lab-2)
   Set Ip Address : 192.168.10.21
   Subnet mask :255.255.255.0
   For **DHCP** setup:-
   - Starting IP: 192.168.10.21
   - Subnet mask : 255.255.255.0
5. Step:5:-

Now , we setup laptop ,Smart phone and PC Like (Lab-2)

Click PC → Desktop → IP Configuration ->Static->IPv4

Since all PCs and laptops use **DHCP**, the server will assign IP addresses sequentially starting from 192.168.10.21

Device Name	IP Address	Subnet Mask
PC1	192.168.10.21	255.255.255.0
Laptop0	192.168.10.22	255.255.255.0
Smartphone0	192.168.10.23	255.255.255.0
PC2	192.168.10.24	255.255.255.0
PC3	192.168.10.25	255.255.255.0
PC4	192.168.10.26	255.255.255.0
PC5	192.168.10.27	255.255.255.0
PC6	192.168.10.28	255.255.255.0

6. Step-6 :-

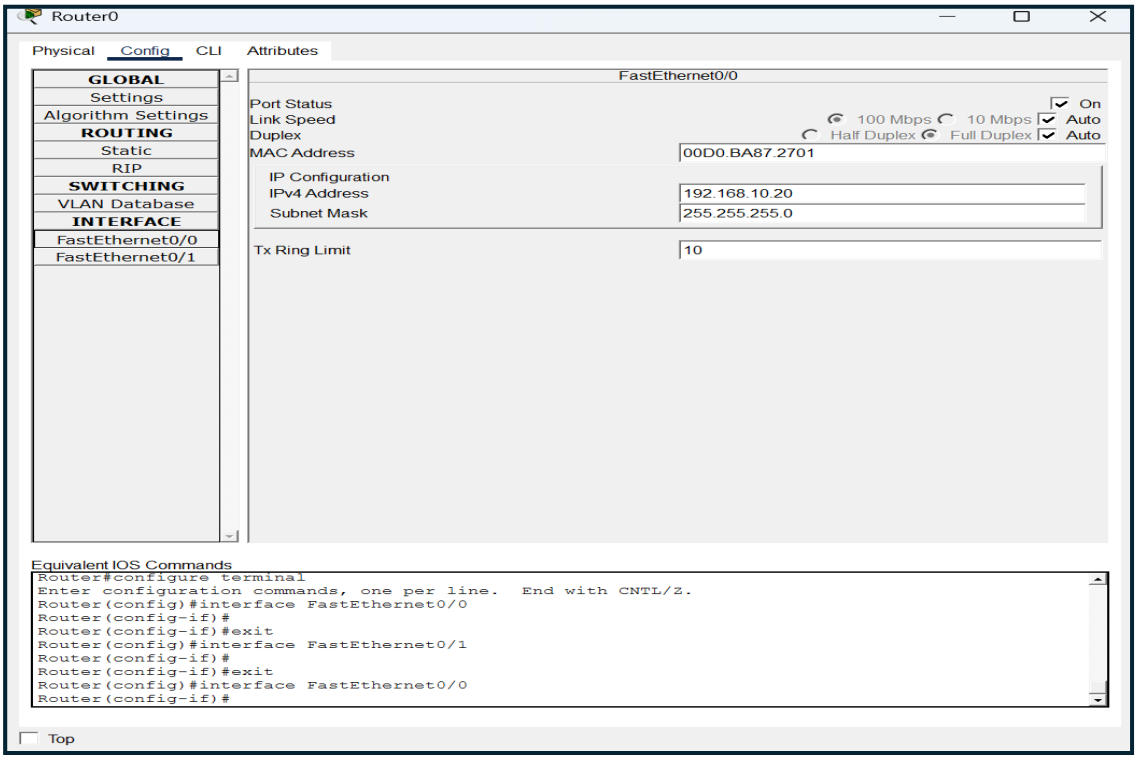
   Now, We setup our Router :-

   Click Router0 → Config → FastEthernet0/0

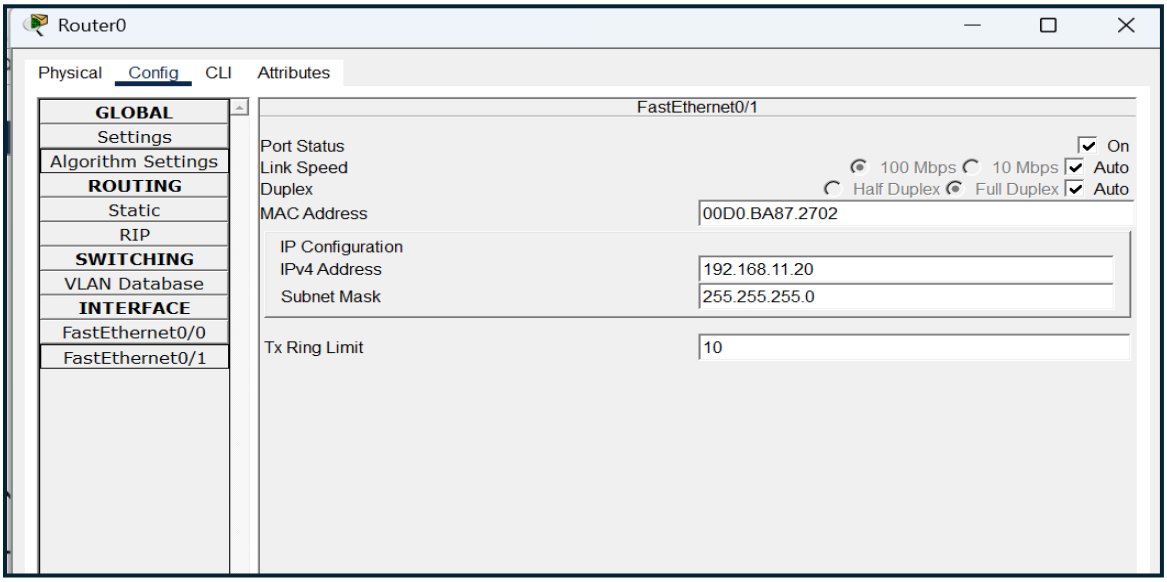
   Set :-

   IP Configuration :192.168.10.20

   Subnet Mask :255.255.255.0



Then, Click Router0 → Config → FastEthernet0/1



Set,

IP Configuration :-192.168.11.20

Subnet Mask :-255.255.255.0

Click Router0 → Config → Static

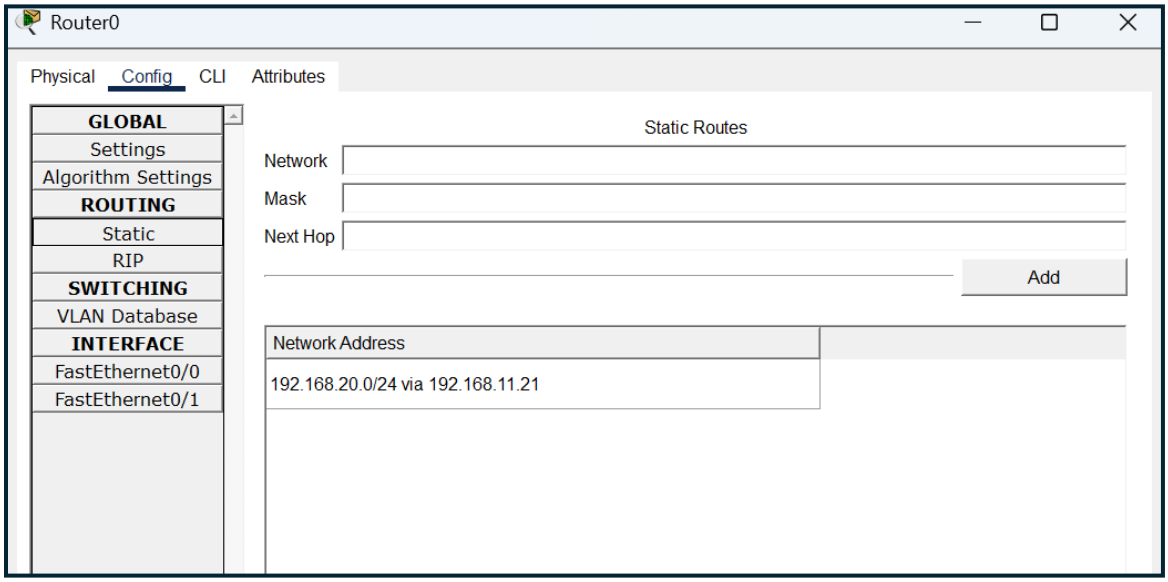
And set ,

Network:-192.168.20.0

Mask:-255.255.255.0

Next Hop :-192.168.11.21

Then Add this configuration.

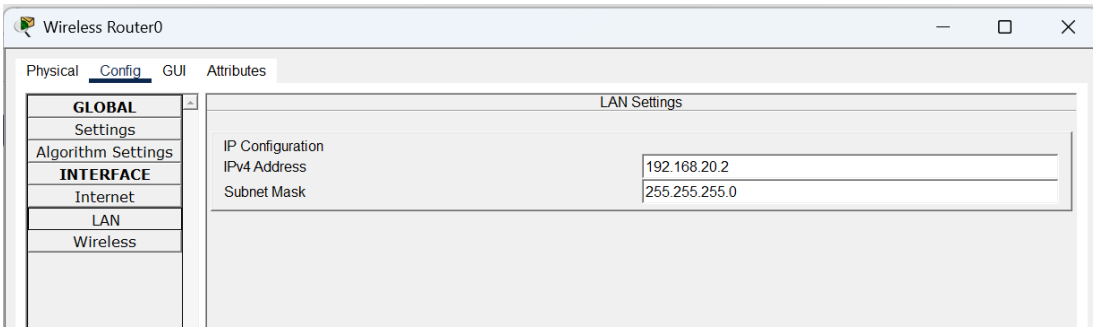


Now , We will configure Router1 in the same way :-

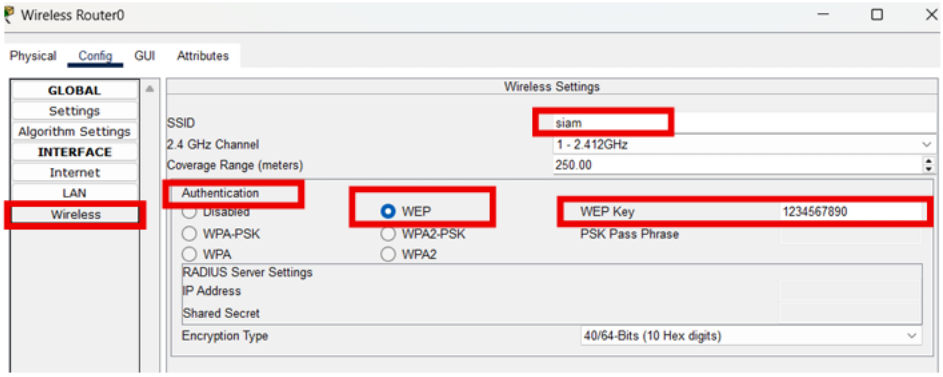
- Click Router0 → Config → FastEthernet0/0
  Set :-
  IP Configuration : 192.168.11.21
  Subnet Mask :255.255.255.0

- Click Router0 → Config → FastEthernet0/1
  Set :-
  IP Configuration : 192.168.20.1
  Subnet Mask :255.255.255.0
- Click Router0 → Config → Static
  Network:-192.168.10.0
  Mask:-255.255.255.0
  Next Hop :-192.168.11.20
  Then Add this configuration

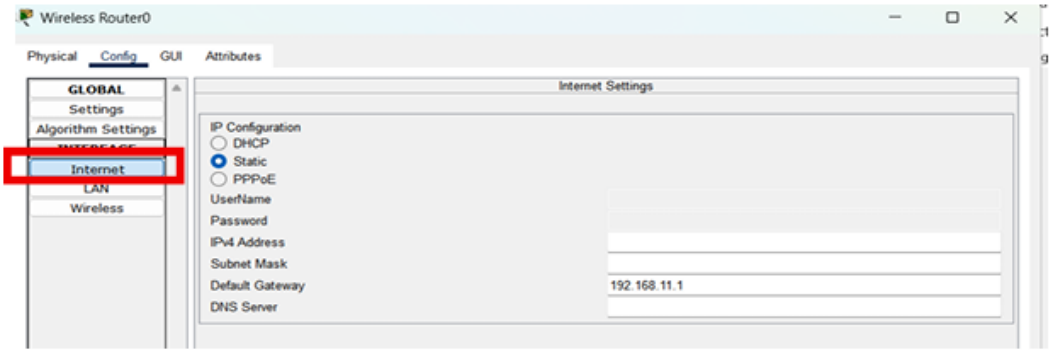
7. Step-7:-
   In this part we configure the wireless router .
   1/Go to Wireless tab:
   • SSID: siam
   • Authentication: WEP
   • Password/WEP key: 1234567890



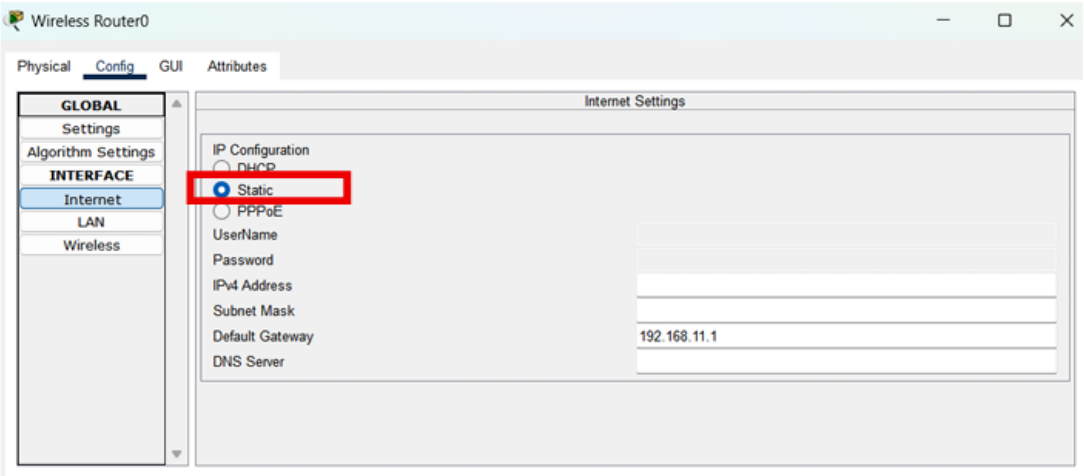
2/ in the Config tab, click on Internet settings.



3/ Set the Internet Connection Type to Static IP



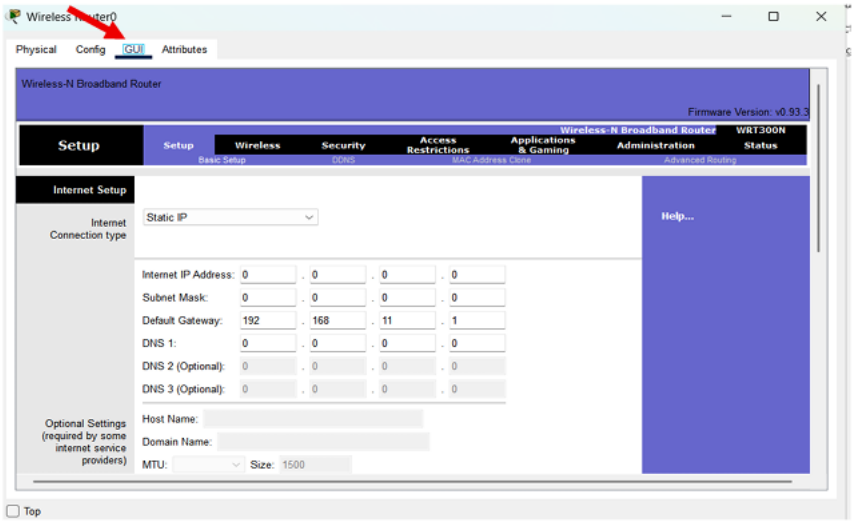
4/ Assign a Default Gateway (it will typically be the IP of Router0 or internet gateway, which is 192.168.11.1).



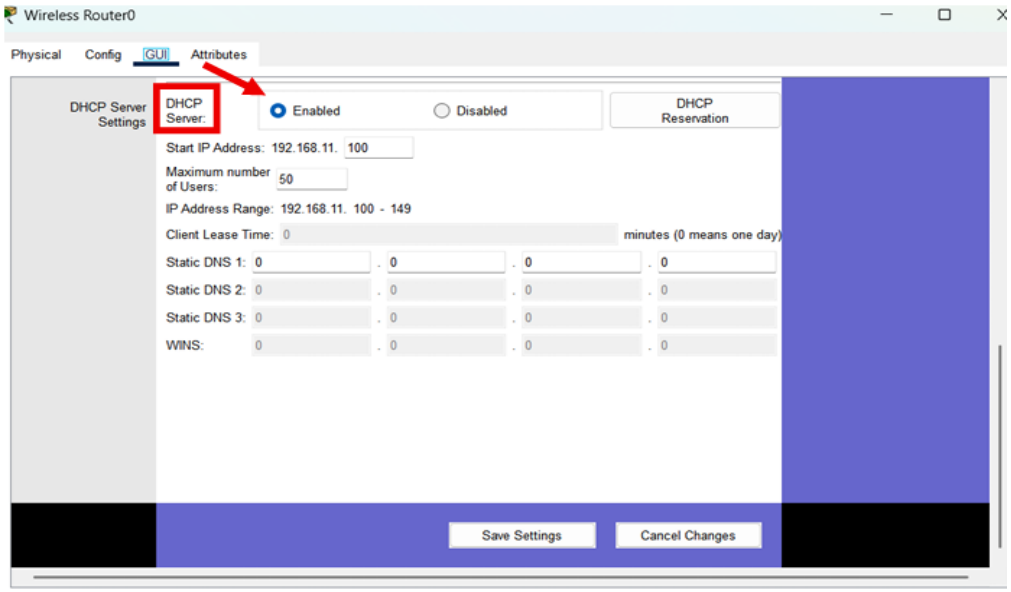
5/ Leave the Internet IP Address and other settings as 0.0.0.0, unless you have a specific internet connection configuration.

Now , **Configure the DHCP Server**

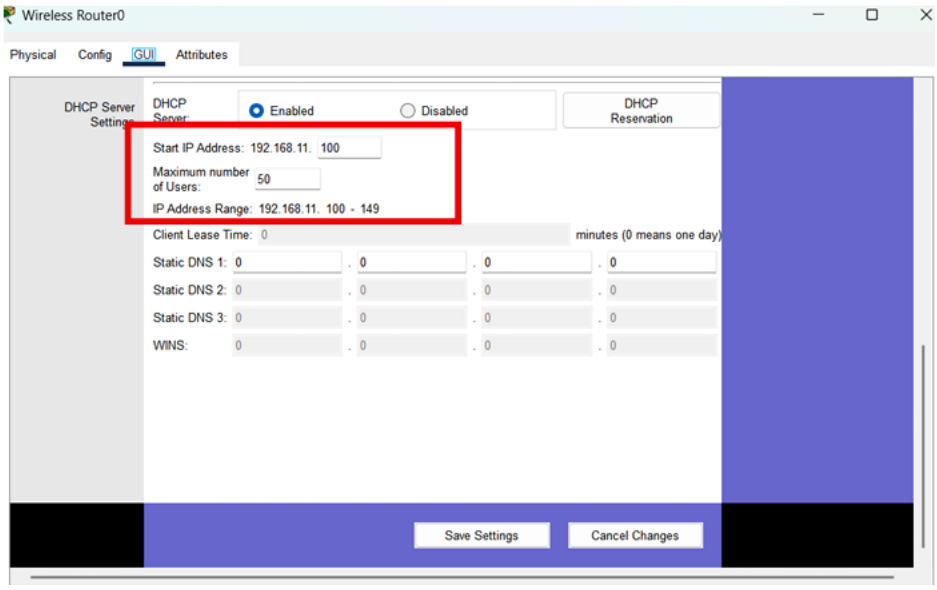
1/Click GUI Tab



2/ Under DHCP Server Settings, enable the DHCP service by clicking on Enabled.



3/ Set up the DHCP Pool:
Start IP Address: 192.168.11.100
End IP Address: 192.168.11.149
Subnet Mask: 255.255.255.0



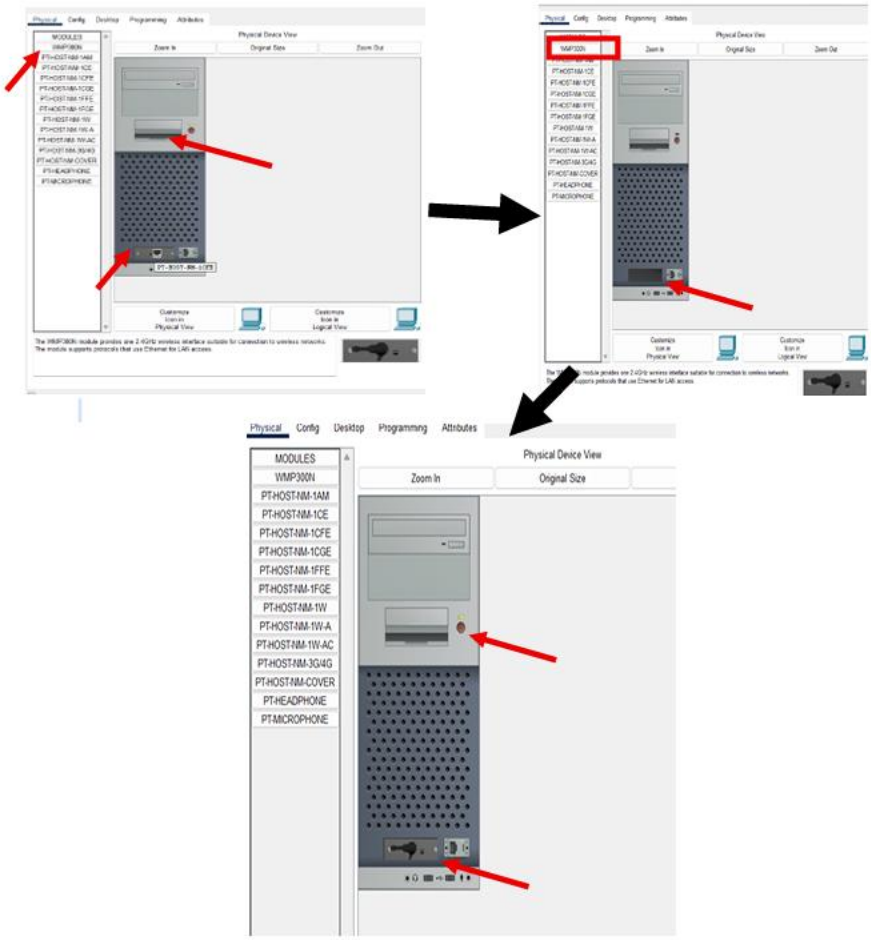
4.close the tab

8. Step-8:-

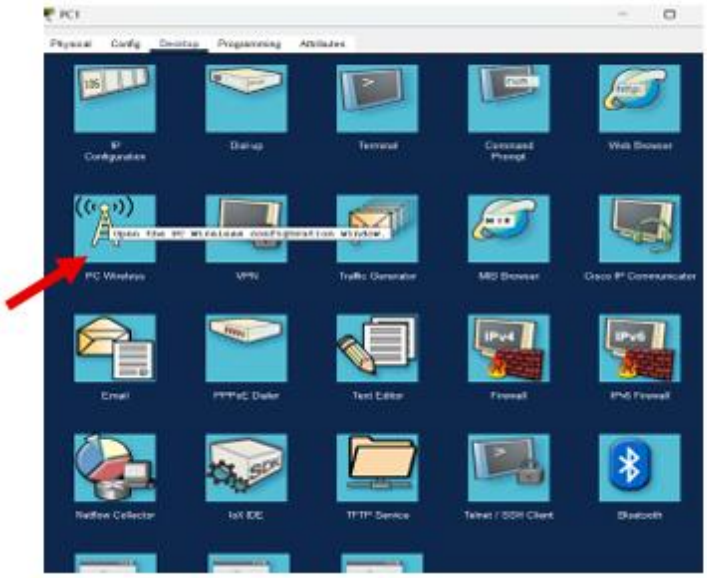
   In, this part we configure our Wireless router , wireless pc ,laptop and smartphone.

   For each Laptop / PC wireless: as like the picture

   1. Click the device->physical->power off module->Drag the port to open->select the wpm 300N and drop to the port->Turn on the pc



2. Go to Desktop → PC Wireless / Laptop Wireless



3. Select the SSID shown

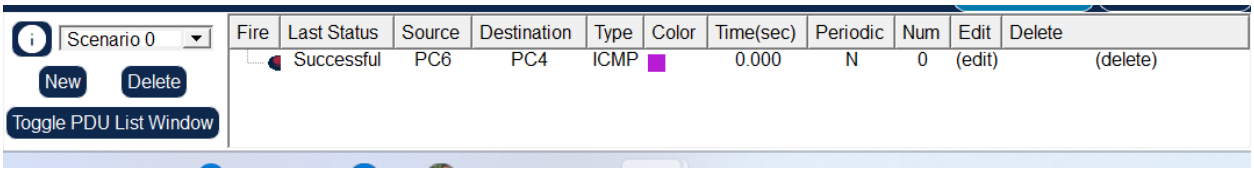


4. Enter the WiFi password-> Click Connect



9.Step-9:-

Transfer a packet as like lab 2



Final picture of completed lab task4:

